

★ WEEK 2 · PART 1 of 3

Stationarity: The Concept

The Gatekeeper Before Every Model

Strict vs weak · autocovariance & ACF · white noise · AR(1) · trend vs difference

Almost every model you will build assumes stationarity — that the statistical rules of the game do not change over time. This part makes that assumption precise, and teaches your eye to spot when it fails.

1

PART 1 · STATIONARITY

When Do the Rules Stay the Same?

From an intuitive idea to two precise definitions, the ACF as a diagnostic, and the AR(1) process where one parameter decides everything.

The Regression That Lied — Part 1

A 1974 simulation that shook econometrics

Granger and Newbold generated two completely UNRELATED random series, regressed one on the other — and kept finding strong, 'highly significant' relationships that did not exist.

The experiment

- Simulate two independent random walks — no link whatsoever.
- Regress one on the other and read off R^2 and the t-statistic.
- Repeat thousands of times and tally the 'significant' results.

The shock

- Standard tests screamed 'significant' the vast majority of the time.
- R^2 was often high; t-statistics were enormous.
- The textbook regression toolkit was quietly lying.

The Regression That Lied — Part 2

The result, and the lesson for us

The culprit was non-stationarity. Regressing trending, non-stationary series makes ordinary t-tests and R^2 wildly unreliable — the 'spurious regression' problem.

Why it matters

- Two series can share a trend and look related when they are not.
- The usual significance tests assume something these series violate.
- Fortunes and papers have been built on such phantom relationships.

The lesson

- Before you regress or model anything, check stationarity FIRST.
- That single habit is the whole point of Week 2.
- We build the mechanism formally in Part 3 — Granger later won a Nobel for the cure.

Why Stationarity Is the Gatekeeper

Three reasons it comes before every other tool

1

Estimation needs a stable target

To estimate a mean, variance, or autocorrelation, that quantity must actually be constant. If it drifts, our estimates chase a moving target.

2

Inference breaks without it

t-tests, R^2 , and standard errors are derived assuming stationarity. Apply them to non-stationary data and they mislead — the spurious-regression trap.

3

Forecasting requires a stable process

A forecast extrapolates the past into the future. That only works if the future obeys the same rules as the past — the essence of stationarity.

Rule: diagnose stationarity before you estimate, test, or forecast anything.

What Is Stationarity?

The statistical 'rules of the game' do not change over time

A time series is stationary if its statistical properties are invariant to a shift in time — the process looks the same whether you observe it now or a decade from now.

The intuition

- Slide a window along the series: the picture should look the same.
- No trend, no drifting spread, no changing cycles.
- Only then is 'the average' or 'the variance' a meaningful number.

Two versions

- Strict: the entire distribution is shift-invariant (very strong).
- Weak: only the first two moments are (the workhorse).
- We define both next, then use weak stationarity throughout.

Strict Stationarity

The strong — often too strong — definition

$$\{X_t, X_{t+1}, \dots, X_{t+k}\} \stackrel{d}{=} \{X_{t+h}, \dots, X_{t+k+h}\} \quad \forall h$$

What it demands

- Every joint distribution is unchanged by a time shift h .
- Not just the mean and variance — the WHOLE distribution.
- Skewness, kurtosis, tail shape: all time-invariant.

Why we rarely use it

- It is extremely hard to verify from a single sample path.
- It is stronger than most models actually need.
- So we work with a weaker, checkable version →

Weak (Covariance) Stationarity

The workhorse definition — three simple conditions

$$\mathbb{E}[X_t] = \mu, \quad \text{Var}(X_t) = \sigma^2, \quad \text{Cov}(X_t, X_{t-k}) = \gamma(k)$$

The three conditions

- 1. Constant mean μ for all t .
- 2. Constant, finite variance σ^2 for all t .
- 3. Autocovariance depends only on the lag k , not on t .

Why this is enough

- It pins down exactly what most models rely on.
- All three are estimable from one sample path.
- From here on, 'stationary' means weakly stationary.

Strict vs Weak Stationarity

When the two coincide

Property	Strict stationarity	Weak (covariance) stationarity
What is invariant	the ENTIRE joint distribution	only mean, variance, autocovariance
Verifiability	very hard from one sample	checkable from one sample
Role	a theoretical ideal	the practical workhorse
When they coincide	—	for a Gaussian process, weak implies strict

Takeaway: for the Gaussian world we usually assume, weak stationarity already gives us strict stationarity.

Autocovariance and the ACF

Measuring how a series relates to its own past

$$\gamma(k) = \text{Cov}(X_t, X_{t-k}), \quad \rho(k) = \frac{\gamma(k)}{\gamma(0)}$$

What they measure

- $\gamma(k)$: the covariance between the series and its k -lagged self.
- $\rho(k)$: the same, scaled to lie in $[-1, 1]$ — the ACF.
- $\rho(0) = 1$ always; the shape of $\rho(k)$ is the fingerprint.

Why it is central

- For a stationary series $\rho(k)$ depends only on the lag k .
- Its decay pattern is our fastest diagnostic.
- It will also drive the ARMA models of Week 3.

White Noise — The Simplest Stationary Process

Zero mean, constant variance, no memory

$$\mathbb{E}[\varepsilon_t] = 0, \quad \text{Var}(\varepsilon_t) = \sigma^2, \quad \text{Cov}(\varepsilon_t, \varepsilon_s) = 0 \quad (t \neq s)$$

Its properties

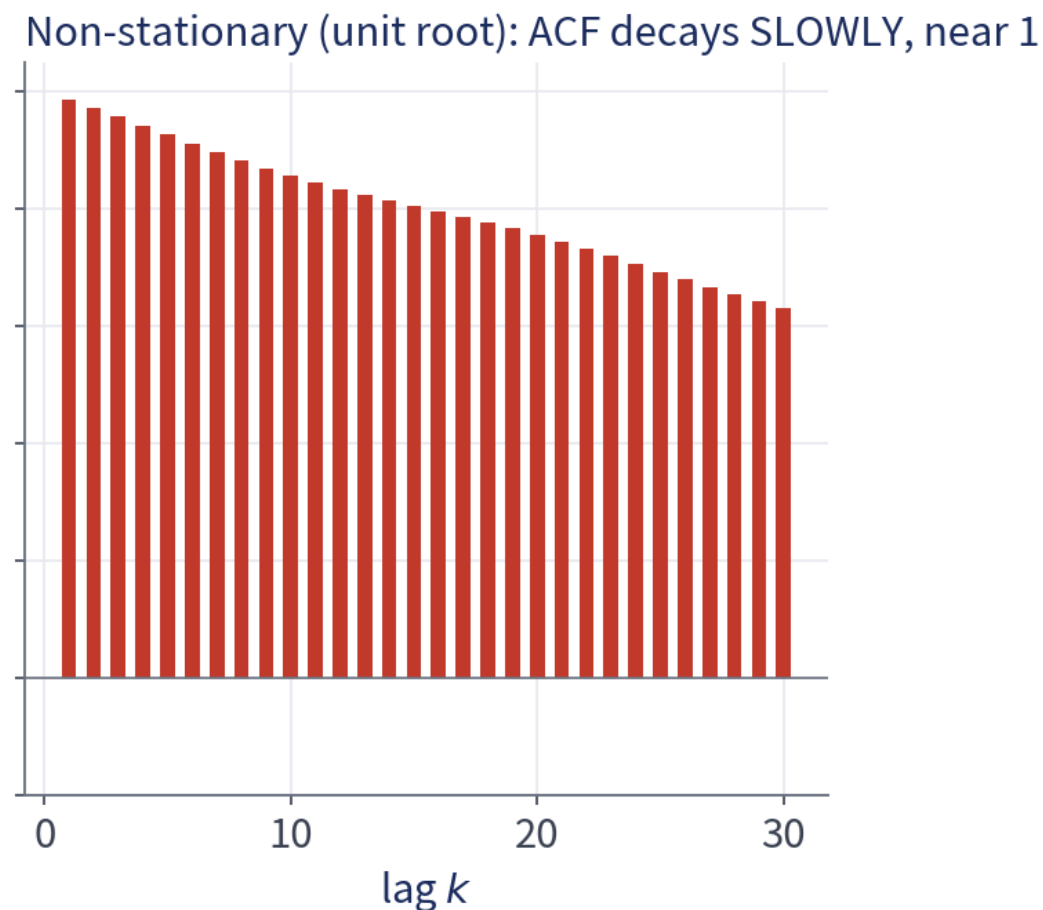
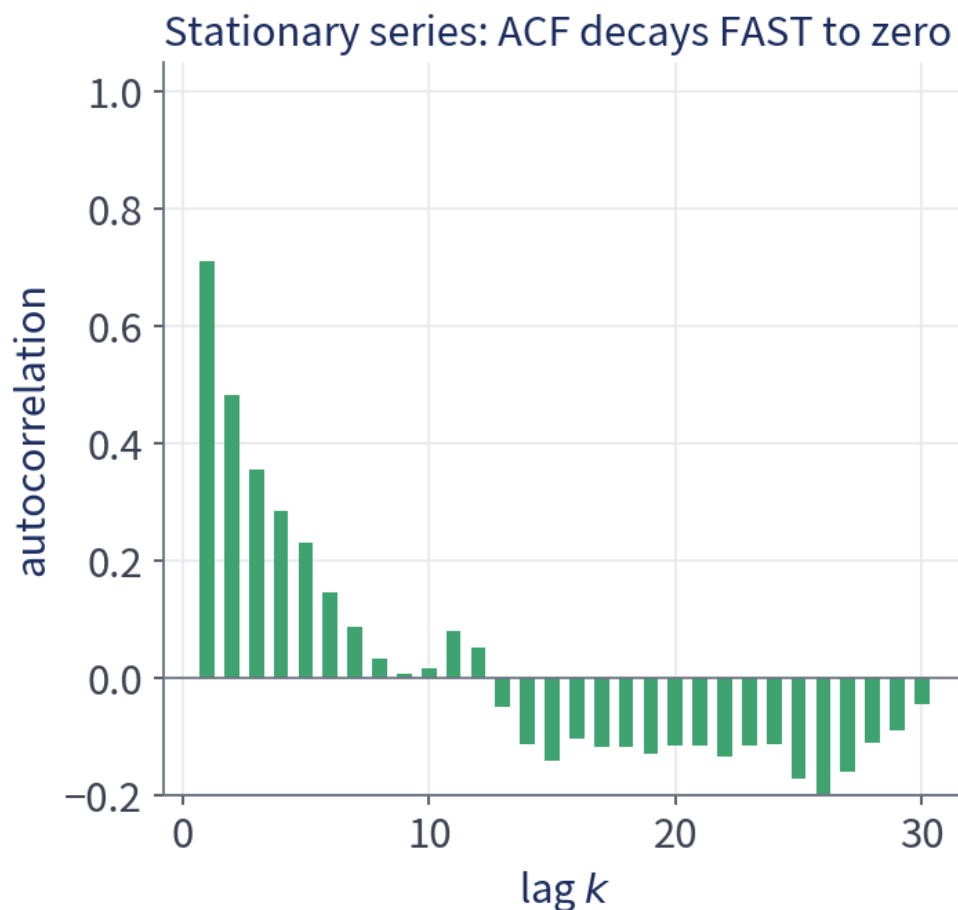
- Uncorrelated over time: $\rho(k) = 0$ for every $k \neq 0$.
- Stationary by construction — the baseline case.
- The 'ideal residual' a good model should leave behind.

Why we care

- It is the building block of every other process.
- AR, MA, and ARMA are all filters applied to white noise.
- If your model's residuals are white noise, you are done.

Reading the ACF: Two Tell-Tale Shapes

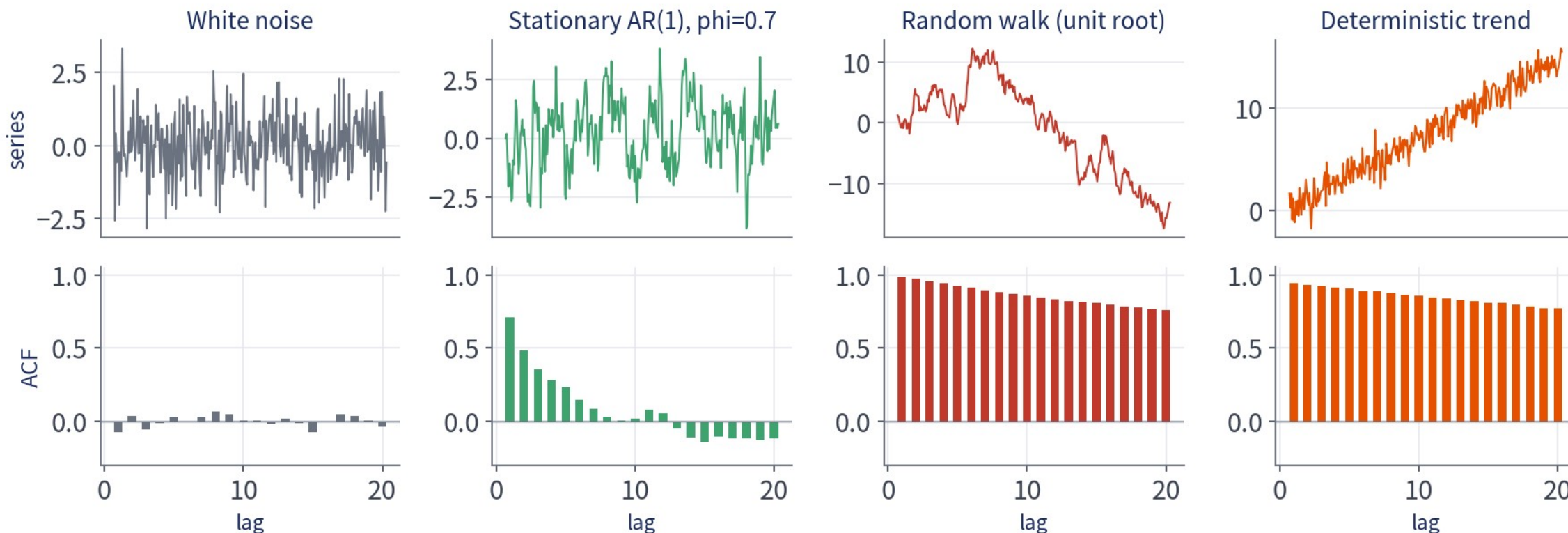
The fastest visual check for non-stationarity. Fast decay (left) says stationary; a slow, near-one decay (right) is the signature of a unit root.



Four Signatures, Side by Side

Train your eye before you trust any test: white noise, a stationary AR(1), a random walk, and a deterministic trend — each with its distinctive series and ACF.

Four signatures: train your eye before trusting any test



The AR(1) Process

One parameter controls persistence

$$X_t = \phi X_{t-1} + \varepsilon_t \quad (\text{AR}(1) \text{ process})$$

How to read it

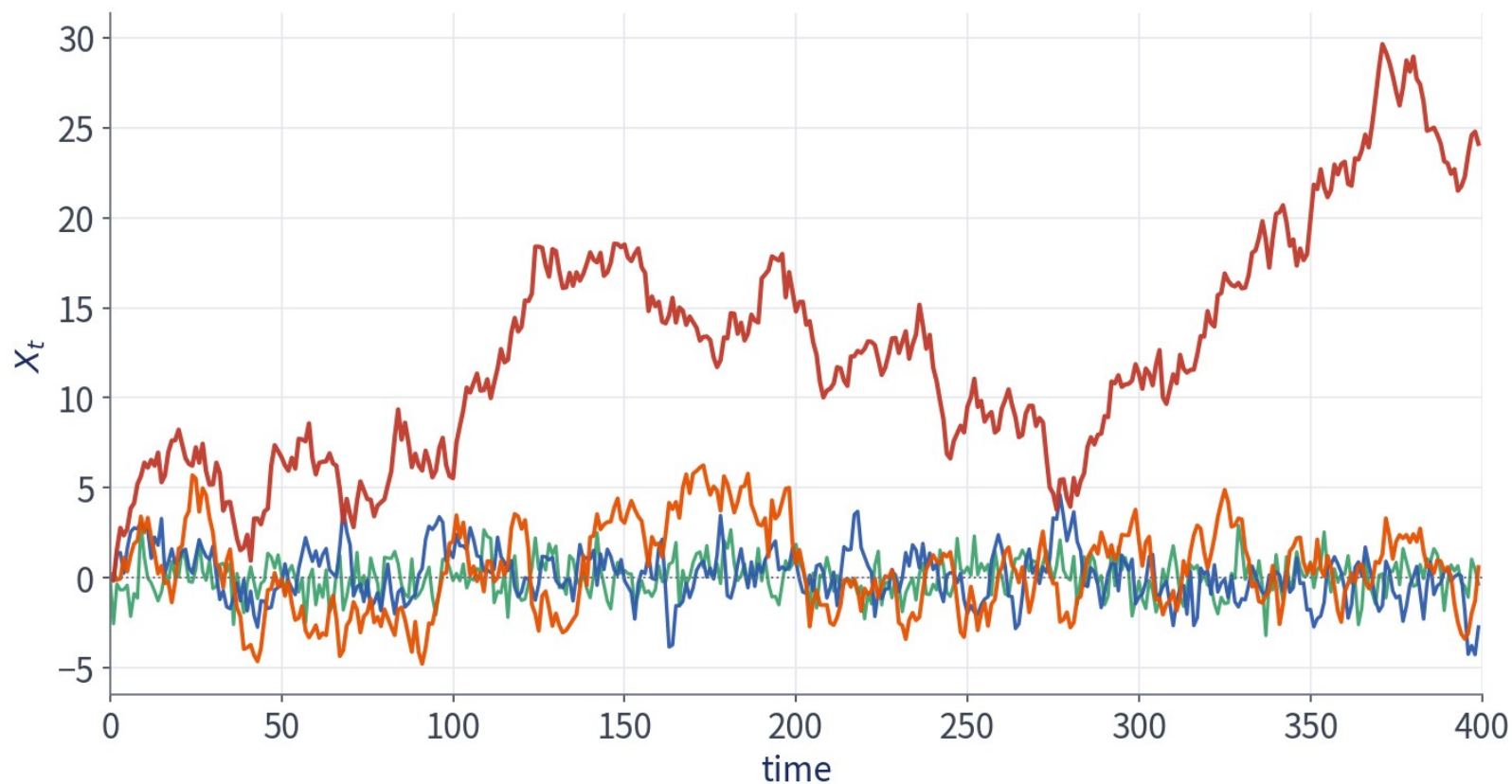
- Today is ϕ times yesterday, plus a fresh shock.
- ϕ measures how strongly the past carries forward.
- $\phi = 0$ gives white noise; ϕ near 1 gives long memory.

Why it is the key example

- It is the simplest process that can be stationary or not.
- The single parameter ϕ decides which.
- It is the cleanest bridge to unit roots (Part 2).

The Role of ϕ in an AR(1)

As ϕ rises toward 1, the series wanders farther from its mean and returns ever more slowly. At $\phi = 1$ it stops returning at all — a random walk.



— $\phi = 0.3$ (weak memory) — $\phi = 0.7$ (moderate) — $\phi = 0.95$ (near unit root) — $\phi = 1.0$ (unit root: random walk)

The Stationarity Condition: $|\phi| < 1$

The line between a stable series and a wandering one

$$|\phi| < 1 \implies \mathbb{E}[X_t] = 0, \quad \text{Var}(X_t) = \frac{\sigma^2}{1-\phi^2}, \quad \rho(k) = \phi^k$$

When $|\phi| < 1$

- The series is stationary with a finite variance.
- Its ACF decays geometrically: $\rho(k) = \phi^k$.
- Shocks fade; the series is mean-reverting.

When $|\phi| = 1$

- The variance is no longer finite — it grows without bound.
- The ACF no longer decays — this is a unit root.
- Shocks are permanent; the series never reverts (Part 2).

Mean Reversion — The Economic Meaning

What $|\phi| < 1$ feels like in practice

In a stationary AR(1), every shock decays: the series is pulled back toward its long-run mean. In a unit-root series, shocks are permanent — there is no anchor to return to.

Stationary (mean-reverting)

- A spread, a valuation ratio, or a real interest rate.
- Deviations tend to be temporary and self-correcting.
- This is what makes some strategies (e.g. pairs trading) possible.

Unit root (no anchor)

- A stock price level or an exchange rate.
- Today's level is the best guess for tomorrow's.
- Betting on 'reversion to a mean' here is a mistake.

Ergodicity — Why One Sample Is Enough

Averaging over time stands in for averaging over outcomes

We only ever see one history of the market. Ergodicity is the extra assumption that lets a single long sample reveal the true underlying averages.

What it says

- Time averages converge to the true population averages.
- The one path we observe is 'representative' of the process.
- Stationarity plus ergodicity makes estimation legitimate.

Why it matters here

- Without it, a sample mean need not tell us the true mean.
- It is the quiet assumption behind every statistic we compute.
- Non-stationary series typically fail it outright.

Trend-Stationary vs Difference-Stationary

Two kinds of non-stationarity need two different cures

$$\underbrace{X_t = a + b t + \varepsilon_t}_{\text{trend-stationary: detrend}}$$

$$\underbrace{\Delta X_t \text{ stationary}}_{\text{difference-stationary}}$$

Trend-stationary

- Stationary around a deterministic trend $a + b \cdot t$.
- The cure is to DETREND — subtract the fitted line.
- Shocks are temporary; the trend is fixed and knowable.

Difference-stationary

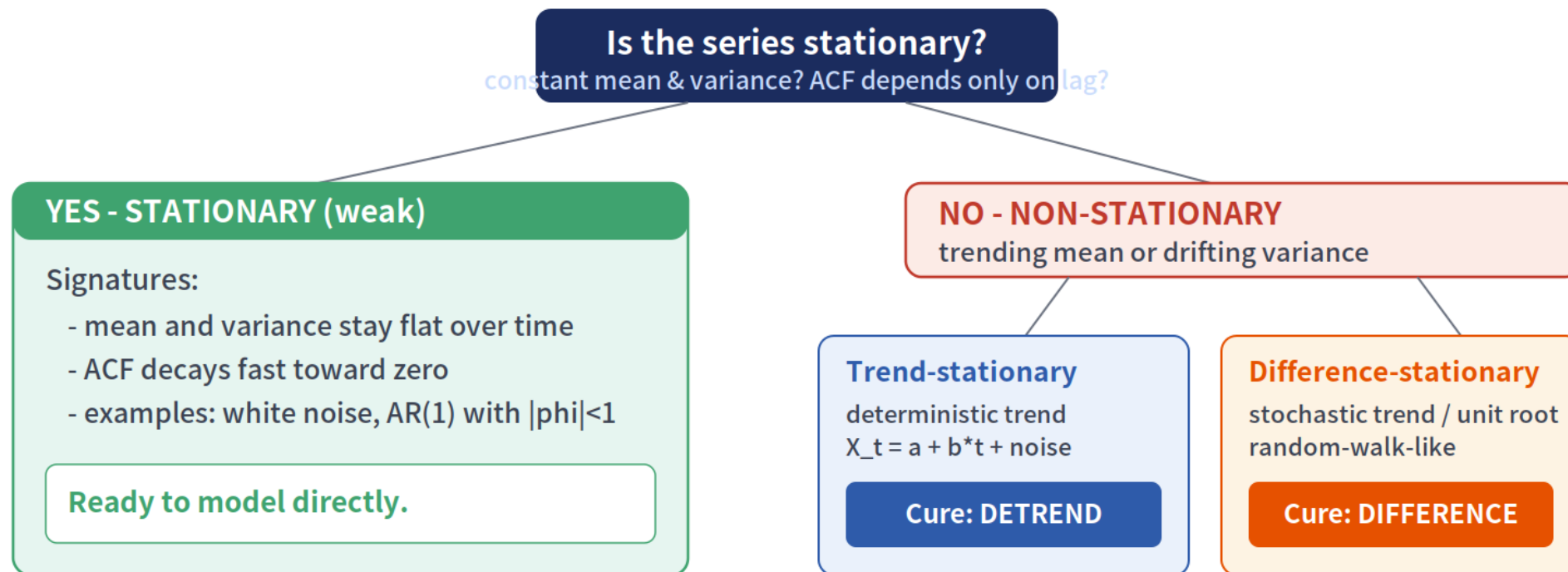
- A stochastic trend — a unit root, random-walk-like.
- The cure is to DIFFERENCE — model ΔX_t instead.
- Shocks are permanent; using the wrong cure ruins the analysis.

The Stationarity Map

One diagnostic question, three destinations, two cures. Detrend a deterministic trend; difference a stochastic one — never mix them up.

The Stationarity Map

One question, three destinations, two cures. Diagnose first, then choose the fix.



A First-Pass Stationarity Checklist

Eyeball it before you formally test it

1 · Plot

the series

Look for a trend, drift, or a changing spread over time.

2 · Split & compare

rolling stats

Do the mean and variance of the first half match the second?

3 · Read the ACF

decay speed

Fast decay hints stationary; slow, near-one decay hints a unit root.

4 · Then test

formalize

Only now run a formal unit-root test (Part 2) to decide.

The eye catches most problems; the formal test settles the close calls.

What Assumes Stationarity (Almost Everything)

Why this gatekeeping matters downstream

Tools that require it

- ARMA models (Week 3): the ACF/PACF only mean something if stationary.
- Regression inference: t-tests and R^2 assume stationary errors.
- Volatility models and most forecasts assume a stable process.

What happens if you skip it

- Spurious regressions: phantom relationships (the 1974 episode).
- Meaningless 'means' and exploding variances.
- Forecasts that extrapolate a trend the data never really had.

Seeing Stationarity

Direct Claude Code — you read and interpret, not hand-code

GOAL Build intuition by generating the four signatures and comparing their series, rolling statistics, and ACFs.

[INPUT] Prompt · Claude Code

```
Write and run stationarity_demo.py that:  
1. simulates four series of length 500 -  
   white noise, an AR(1) with  $\phi=0.7$ ,  
   a random walk, and a linear trend+noise;  
2. for each, plots the series, its rolling  
   mean & std (window 50), and its ACF;  
3. prints the first-half vs second-half mean  
   and variance for each series.  
Lay the plots out in a 4-row grid and tell me,  
for each, which stationarity condition it  
satisfies or breaks.
```

How to work with it

- Match each plot to the four signatures.
- Watch the random walk's rolling stats drift.
- Change ϕ to 0.95 and 1.0 — what changes?
- Keep this script; Part 2 adds the formal tests.

Deliverable: stationarity_demo.py + the 4-row figure you can read at a glance.

Three Questions Before You Move On

If you can answer these, Part 1 has landed

Q1

A series' variance grows steadily over time. Can it be weakly stationary?

No — var not constant

Q2

An AR(1) has $\phi = 0.5$. Stationary or not, and does it mean-revert?

Stationary; reverts

Q3

Your series has a deterministic trend. Difference it or detrend it?

Detrend

All three come straight from the definitions and the AR(1) lens.

Three Misconceptions to Avoid

Where intuition slips

Common myths

- "Stationary means constant / flat."
- "No visible trend means it must be stationary."
- "Differencing is always the right fix."

The reality

- Stationary means stable STATISTICS, not a flat line — it still fluctuates.
- Variance drift or slow ACF decay can hide with no obvious trend.
- Detrend a deterministic trend; difference only a stochastic one.

Three Things to Carry Forward

Part 1 in one screen

1

Stationarity means stable statistics over time

Weak stationarity — constant mean, constant variance, lag-only autocovariance — is the workhorse assumption behind nearly every model.

2

The ACF and the AR(1) are your first diagnostics

Fast ACF decay and $|\phi| < 1$ mean stationary and mean-reverting; slow decay and $\phi = 1$ signal a unit root.

3

Two non-stationarities, two cures

Detrend a deterministic trend; difference a stochastic one. Getting this wrong is the root of the spurious-regression trap.

Next — Part 2: Unit Roots and the formal tests (ADF, PP, KPSS) that settle the diagnosis.

★ WEEK 2 · PART 1 · PRACTICE

Stationarity, Worked by Hand

Check the Three Conditions, Then Find the $|\phi| < 1$ Line

Weak stationarity, the AR(1) boundary, and two cures

Part 1's gatekeeper concept, with the arithmetic shown. We check the three conditions of weak stationarity - watching a random walk's variance grow while white noise and a stable AR stay constant - then locate the $|\phi| < 1$ boundary and match each non-stationarity to its cure.

1P

PART 1 · APPLICATIONS

The Gate Every Model Must Pass

Case A: check weak stationarity's three conditions and watch a random walk fail as its variance grows with time.
Case B: locate the $|\phi| < 1$ boundary between a mean-reverting series and a wandering one, and match trend- and difference-stationarity to their cures. Then a Claude Code lab.

What Part 1 Told Us

Now we apply it - with numbers

1

Stationarity is the gatekeeper

The statistical rules of the game must not change over time - constant mean, constant variance, and lag-only autocovariance.

2

$|\phi| < 1$ separates stable from wandering

An AR(1) is stationary only when its coefficient is below one in magnitude. At one, it becomes a random walk; beyond, explosive.

3

Two non-stationarities, two cures

A deterministic trend is cured by detrending; a stochastic trend, a unit root, is cured by differencing. Using the wrong one fails.

Everything here is a calculator exercise: one variance check, one boundary, one pair of cures.

How These Cases Work

Data, then arithmetic, then meaning

Each case has three steps

- Start from a candidate process.
- Check the conditions, or the boundary.
- Then read stationarity.

What you should watch

- How the random walk's variance grows.
- How $|\phi| = 1$ marks the boundary.
- How each cure matches its trend.

The Three Conditions

Checked on three processes

Weak stationarity asks three things: a constant mean, a constant variance, and an autocovariance that depends only on the lag. We test three processes - white noise, a random walk, and a stable AR(1) - and find that the random walk fails on the second condition: its variance grows without bound as time passes

The three processes

- White noise: pure random shocks.
- Random walk: each step adds a shock.
- AR(1): a fraction of yesterday plus a shock.

What we'll do by hand

- Compute the variance of each over time.
- See which stays constant.
- Identify the one that fails.

Weak Stationarity

Three simple conditions

weak stationarity: (1) $\mathbb{E}[y_t] = \mu$, (2) $\text{Var}(y_t) = \gamma_0$, (3) $\text{Cov}(y_t, y_{t-k}) = \gamma_k$ (lag only)

The three conditions

- The mean is the same at every time.
- The variance is the same at every time.
- Autocovariance depends only on the gap.

Why it's the workhorse

- Strict stationarity is often too strong.
- These three cover what estimation needs.
- Break any one, and the tools fail.

The Variance Over Time

White noise, random walk, AR(1)

time t	1	10	50	100	verdict
white noise	1	1	1	1	constant
AR(1)	1.56	1.56	1.56	1.56	constant
random walk	1	10	50	100	grows!

White noise and the AR(1) hold a constant variance at every time - they pass. The random walk's variance is time itself, climbing from 1 to 100 - it fails the second condition. Growing variance is the classic mark of non-stationarity.

Why the Random Walk Fails

Variance equals time

random walk: $\text{Var}(y_t) = t \sigma^2 : 1, 10, 100, \dots \nearrow \infty \Rightarrow$ variance not constant

The reason

- Each step adds an independent shock.
- Variances add, so they accumulate.
- After t steps, the variance is t .

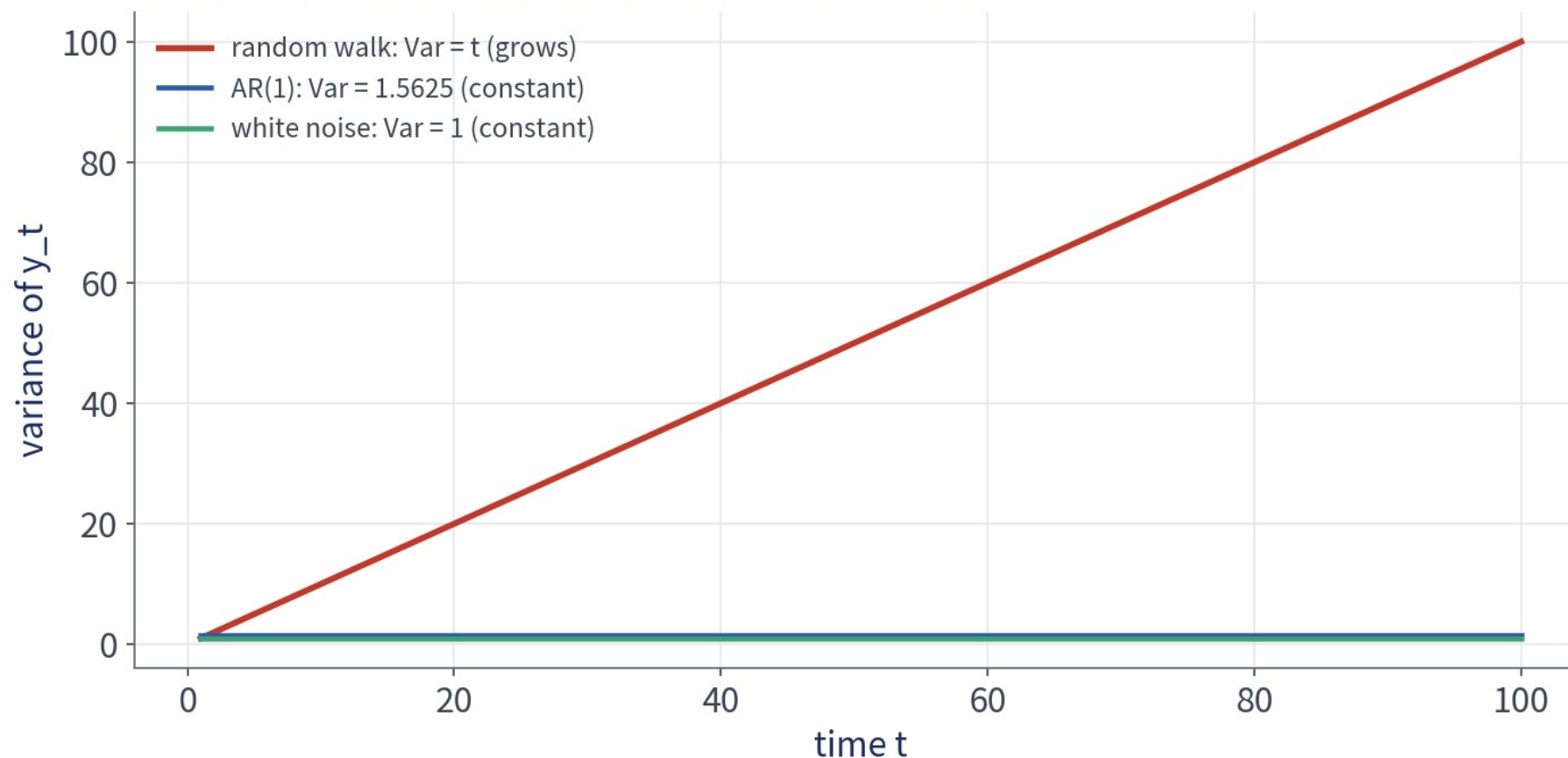
The contrast

- The AR(1) variance settles at 1.5625.
- Because the coefficient is below one.
- The past decays instead of accumulating.

One Line That Won't Stay Flat

White noise and the AR(1) hold flat variance lines - the rules of the game don't change. The random walk's variance climbs straight up, equal to time. That single diverging line is why a random walk fails stationarity, and why prices which behave this way must be transformed before modeling

Case A: only the random walk's variance grows - it fails stationarity



Why the Gate Matters

And how one sample suffices

Why stationarity first

- Estimates only make sense if the rules hold still.
- A growing variance breaks every standard tool.
- So we check it before anything else.

Ergodicity

- For a stationary series, time averages work.
- One long sample stands in for many outcomes.
- That's what lets us estimate at all.

The Boundary and the Cures

$|\phi| < 1$ and $\phi = 1$ and $\phi > 1$ are two cases
An AR(1) is stationary only when its coefficient is below one in magnitude. We compute the variance at three values - 0.6, 1.0, and 1.05 - and watch it go from finite to infinite to undefined, marking the boundary between a mean-reverting series, a random walk, and an explosive one. Then we match each kind of non-stationarity

The three regimes

- Below one: stationary, mean-reverting.
- Exactly one: a random walk.
- Above one: explosive.

What we'll do by hand

- Compute the variance at each coefficient.
- See where it breaks.
- Match trend and difference cures.

The Stationarity Condition

The line at $|\phi| = 1$

$\phi=0.6 : \frac{1}{0.64}=1.56$ (stationary) | $\phi=1 : \frac{1}{0}=\infty$ (random walk) | $\phi=1.05 : \text{explosive}$

The formula

- The variance is the shock variance over one minus phi-squared.
- It's finite only when phi is below one.
- That single condition is the whole test.

The boundary

- At phi below one, it settles.
- At exactly one, it diverges.
- Above one, it has no sensible value.

Three Regimes

Stationary, random walk, explosive

phi	1 - phi-squared	variance	behavior
0.6	0.64	1.5625	stationary, reverts
1.0	0	infinite	random walk
1.05	-0.10	undefined	explosive

At 0.6 the variance is a finite 1.5625, so the series mean-reverts. At exactly 1.0 the denominator is zero and the variance is infinite - a random walk. Above 1.0 it turns negative, meaningless: the process explodes, with shocks amplifying instead of decaying.

Two Cures for Two Trends

Detrend or difference

trend-stationary $\xrightarrow{\text{detrend}}$ stationary; difference-stationary $\xrightarrow{\text{difference}}$ stationary

Trend-stationary

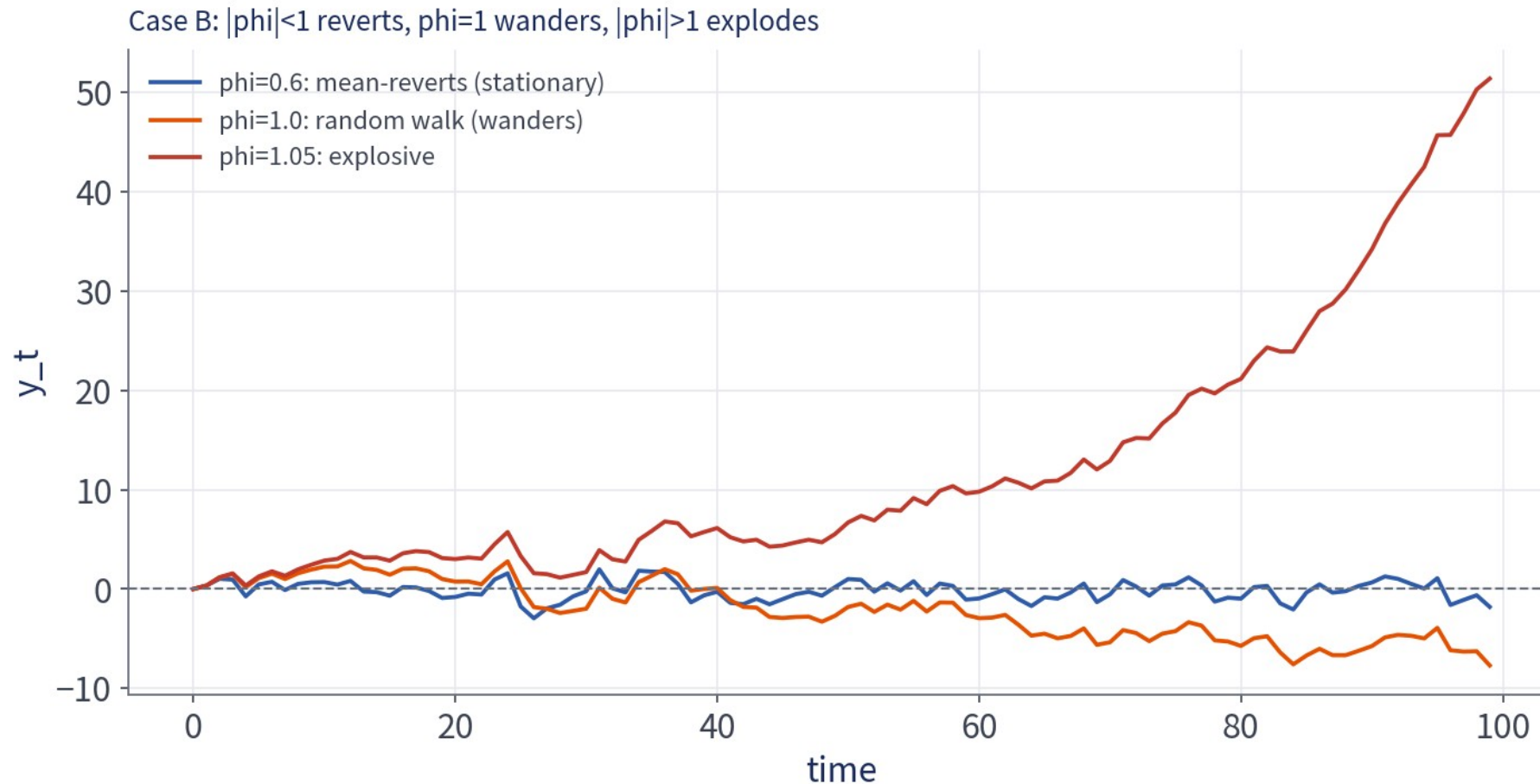
- A deterministic trend plus stationary noise.
- Cure it by detrending - regress out the trend.
- What's left is stationary.

Difference-stationary

- A stochastic trend - a unit root.
- Cure it by differencing.
- Prices are this kind; returns are the fix.

Reverts, Wanders, Explodes

Three AR(1) paths at different coefficients. At 0.6 (blue) the series is pulled back to zero again and again - stationary. At 1.0 (orange) it wanders freely with no anchor - a random walk. At 1.05 (red) it races off to infinity - explosive. The coefficient alone decides which world you're in



Mean Reversion, Made Concrete

What the boundary feels like

Below the boundary

- Shocks decay - 60% remains after one step.
- The series is pulled back to its mean.
- Half a shock is gone in about 1.4 lags.

At and above it

- At one, shocks are permanent - no reversion.
- Above one, shocks amplify and explode.
- The wrong cure leaves the trend behind.

The Stationarity Check

Before modeling at all

Stationarity is the first question you ask of any series. Check the three conditions - especially whether the variance stays constant. For an AR, confirm the coefficient is below one. And when a series isn't stationary, diagnose which kind: a deterministic trend calls for detrending, a stochastic one for differencing. Get this gate right,

or

The conditions (Case A)

- White noise and AR(1): constant variance.
- Random walk: variance equals time.
- Only the first two are stationary.

The boundary (Case B)

- ϕ 0.6 reverts, 1.0 wanders, 1.05 explodes.
- Detrend a trend, difference a unit root.
- The right cure matters.

Reproduce the Variance and the Boundary

Direct Claude Code to check your arithmetic

GOAL Show the random walk's growing variance and the AR(1) $|\phi| < 1$ boundary.

[INPUT] Prompt · Claude Code

Write and run part1_practice.py that:

CASE A - variance over time:

- * simulate white noise, a random walk, and an AR(1) with $\phi=0.6$; print the variance at $t=1, 10, 50, 100$;
- * confirm only the random walk's grows.

CASE B - the boundary:

- * for $\phi=0.6, 1.0, 1.05$, print $\sigma^2/(1-\phi^2)$ and label stationary / random walk / explosive;
- * simulate one path of each.

Then plot the variances and the three paths.

Check against the slides

- random walk $\text{Var} = t$ (1, 10, 100).
- AR(1) $\text{Var} = 1.5625$.
- $\phi=1$ diverges, 1.05 explodes.
- detrend vs difference.

Deliverable: part1_practice.py whose printed numbers match the variances and the boundary.

Three Questions on the Cases

If you can answer these, the practice has landed

Q1 Which condition does a random walk break?

**Constant variance -
its variance grows
with time**

Q2 When is an AR(1) stationary?

**When the coefficient
is below one in
magnitude**

Q3 How do the two cures differ?

**Detrend a
deterministic trend,
difference a unit root**

All three come from the variance check, the $|\phi| < 1$ boundary, and the two cures.

Three Pitfalls to Avoid

Where stationarity trips people

Common mistakes

- "A constant mean is enough for stationarity."
- " ϕ just below one is basically stationary."
- "Any non-stationary series should be differenced."

The reality

- The variance must be constant too - watch it.
- Near one means near-permanent shocks - be careful.
- Detrend a deterministic trend; don't over-difference.

Two Calculations to Remember

Part 1 applications in one screen

1

The random walk fails on variance

White noise and a stable AR(1) held a constant variance - 1 and 1.5625 - at every time, passing the second condition. The random walk's variance was time itself, climbing from 1 to 100. That growing variance is the signature of non-stationarity.

2

The boundary sits at $|\phi| = 1$

At ϕ of 0.6 the AR(1) variance was a finite 1.5625, so it mean-reverts. At exactly 1.0 it was infinite - a random walk - and at 1.05 undefined, an explosive process. The single coefficient decides which regime you're in.

3

Match the cure to the trend

A deterministic trend is cured by detrending; a stochastic trend, a unit root, by differencing. Prices are difference-stationary, which is why we model returns. Getting the diagnosis right is the whole game.

Next — Part 2 practice: unit-root tests, formally distinguishing a random walk from a stationary series.

★ WEEK 2 · PART 2 of 3

Unit Roots and Their Tests

From a Diagnosis to a Formal Decision

Random walks · orders of integration · ADF, PP, KPSS · differencing

Part 1 trained your eye. Part 2 gives you the formal machinery: what a unit root is, why it is dangerous, and the tests that decide — objectively — whether a series must be differenced.

2

PART 2 · UNIT ROOTS

Testing for Non-Stationarity

The unit root through the AR(1) lens, the ADF and KPSS tests that detect it, and the right cure once you have — differencing, done carefully.

From Eyeballing to Testing

Part 1 gave intuition; Part 2 gives a verdict

1

The eye is fast but fallible

A slow-decaying ACF hints at a unit root, but 'slow' is subjective. Two analysts can look at the same plot and disagree.

2

We need an objective, repeatable test

A formal hypothesis test turns the picture into a decision with a stated error rate — the Augmented Dickey-Fuller test and its companions.

3

And then the right cure

If a unit root is present, we difference; if a deterministic trend, we detrend. Choosing correctly is the whole game.

Part 2 in one line: is there a unit root — and if so, difference it.

The Unit Root, Through the AR(1) Lens

One parameter crosses a line — and everything changes

$$X_t = \phi X_{t-1} + \varepsilon_t, \quad \text{unit root} \iff \phi = 1$$

What 'unit root' means

- The AR(1) root sits exactly at 1 when $\phi = 1$.
- Shocks no longer decay — they accumulate forever.
- The variance grows without bound; there is no fixed mean.

Why it is the boundary

- $|\phi| < 1$: stationary, mean-reverting, finite variance.
- $\phi = 1$: a unit root — the knife's edge into non-stationarity.
- This single point is what every test is trying to detect.

The Random Walk Is the Unit Root

The canonical non-stationary process — with or without drift

$$X_t = \mu + X_{t-1} + \varepsilon_t \quad (\text{random walk with drift } \mu)$$

Reading it

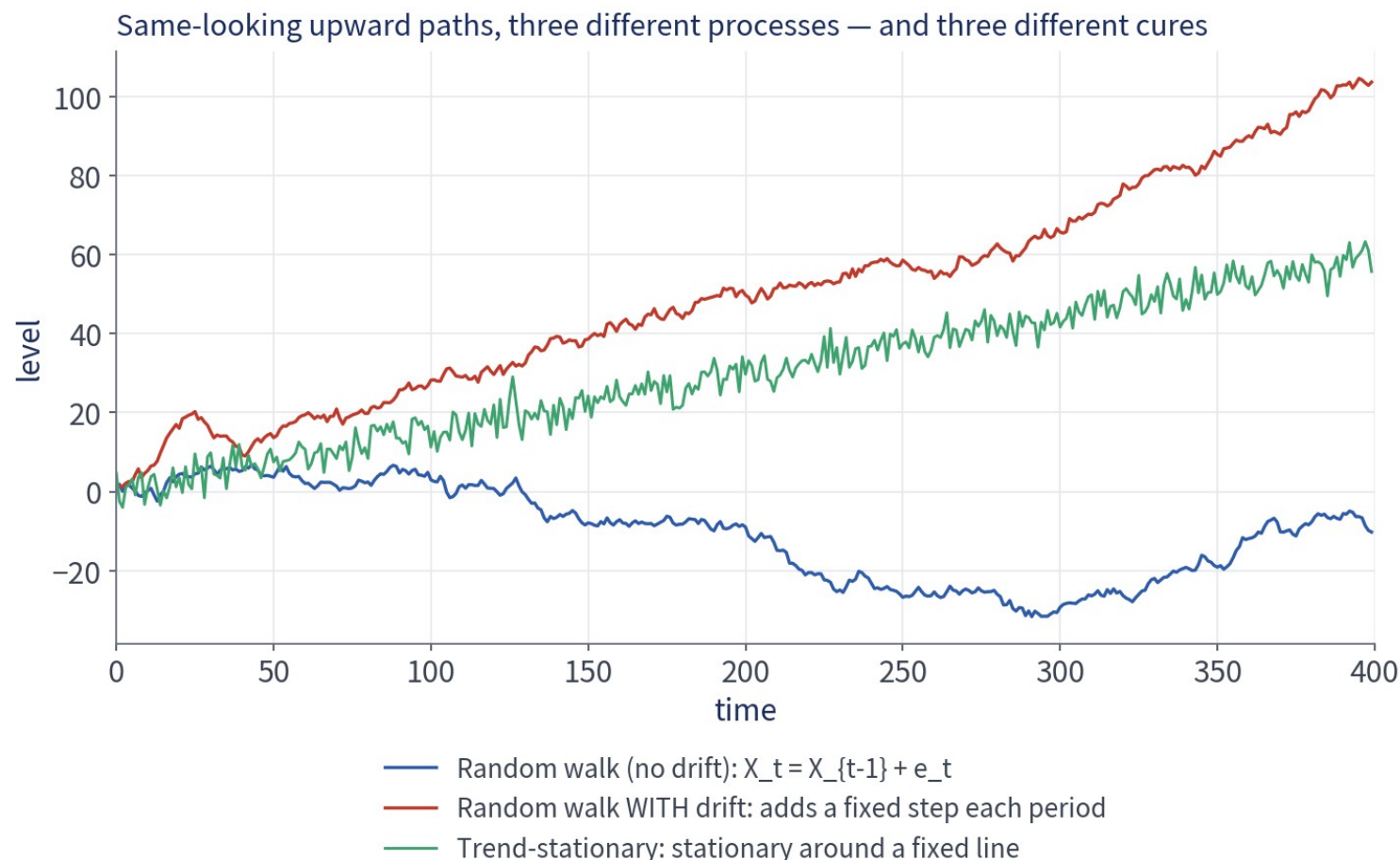
- Set $\phi = 1$ in the AR(1) and you get a random walk.
- Add a constant μ and it drifts steadily up or down.
- The drift is a stochastic trend, not a deterministic one.

Why this matters

- Prices behave like random walks with small drift.
- A drift can masquerade as a genuine trend in a plot.
- Only a test — or differencing — tells them apart.

Three Upward Paths, Three Different Processes

A random walk, a random walk with drift, and a trend-stationary series can all look alike — yet a unit-root test and the correct cure differ for each.



Orders of Integration: I(0), I(1), I(2)

How many differences make a series stationary?

$$X_t \sim I(d) \iff \Delta^d X_t \text{ is stationary,} \quad \Delta X_t = X_t - X_{t-1}$$

The ladder

- I(0): already stationary — no differencing needed.
- I(1): stationary after one difference (most price levels).
- I(2): needs two differences (rare — some price indices).

Why we label it

- The order tells you exactly how to make the series stationary.
- Returns are the first difference of log prices $\rightarrow$ I(0).
- Getting the order wrong means under- or over-differencing.

What Breaks Under Non-Stationarity

Why the diagnosis is not optional

What breaks	Why it fails	The tell-tale sign
Sample mean & variance	No fixed value to estimate; they drift	Rolling statistics never settle
t-tests & standard errors	Derived assuming stationarity	Wildly overstated significance
R-squared	Inflated by shared trends	High R^2 on unrelated series
Forecasts	Extrapolate a phantom trend	Confidence bands explode outward

Every one of these failures traces back to a single unchecked assumption: stationarity.

From a Picture to a Formal Decision

How a unit-root test is actually set up

A unit-root test rewrites the AR model in terms of the CHANGE in the series, then asks a single question: is the coefficient on the lagged LEVEL zero?

The clever trick

- Rewrite so the key parameter is $\gamma = \phi - 1$.
- A unit root ($\phi = 1$) becomes simply $\gamma = 0$.
- So we test one coefficient against zero.

Reading the result

- $\gamma < 0 \rightarrow$ the series pulls back $\rightarrow$ stationary.
- $\gamma = 0 \rightarrow$ no pull-back $\rightarrow$ a unit root.
- The catch is that this test needs a special distribution.

The Dickey-Fuller Test

The original unit-root test

$$\Delta X_t = \alpha + \beta t + \gamma X_{t-1} + \varepsilon_t, \quad H_0 : \gamma = 0 \text{ (unit root)}$$

The regression

- Regress the change ΔX_t on the lagged level X_{t-1} .
- Optionally include a constant and a time trend.
- The t-statistic on X_{t-1} is the test statistic.

The hypotheses

- $H_0: \gamma = 0$ — a unit root is present (non-stationary).
- $H_1: \gamma < 0$ — the series is stationary.
- A sufficiently negative statistic rejects the unit root.

The Augmented Dickey-Fuller (ADF) Test

The workhorse — Dickey-Fuller, with lags added

$$\Delta X_t = \alpha + \beta t + \gamma X_{t-1} + \sum_{j=1}^p \delta_j \Delta X_{t-j} + \varepsilon_t$$

What the lags fix

- Real series have their own short-run autocorrelation.
- Extra lagged differences soak that up.
- This keeps the test valid for realistic data.

Using it in practice

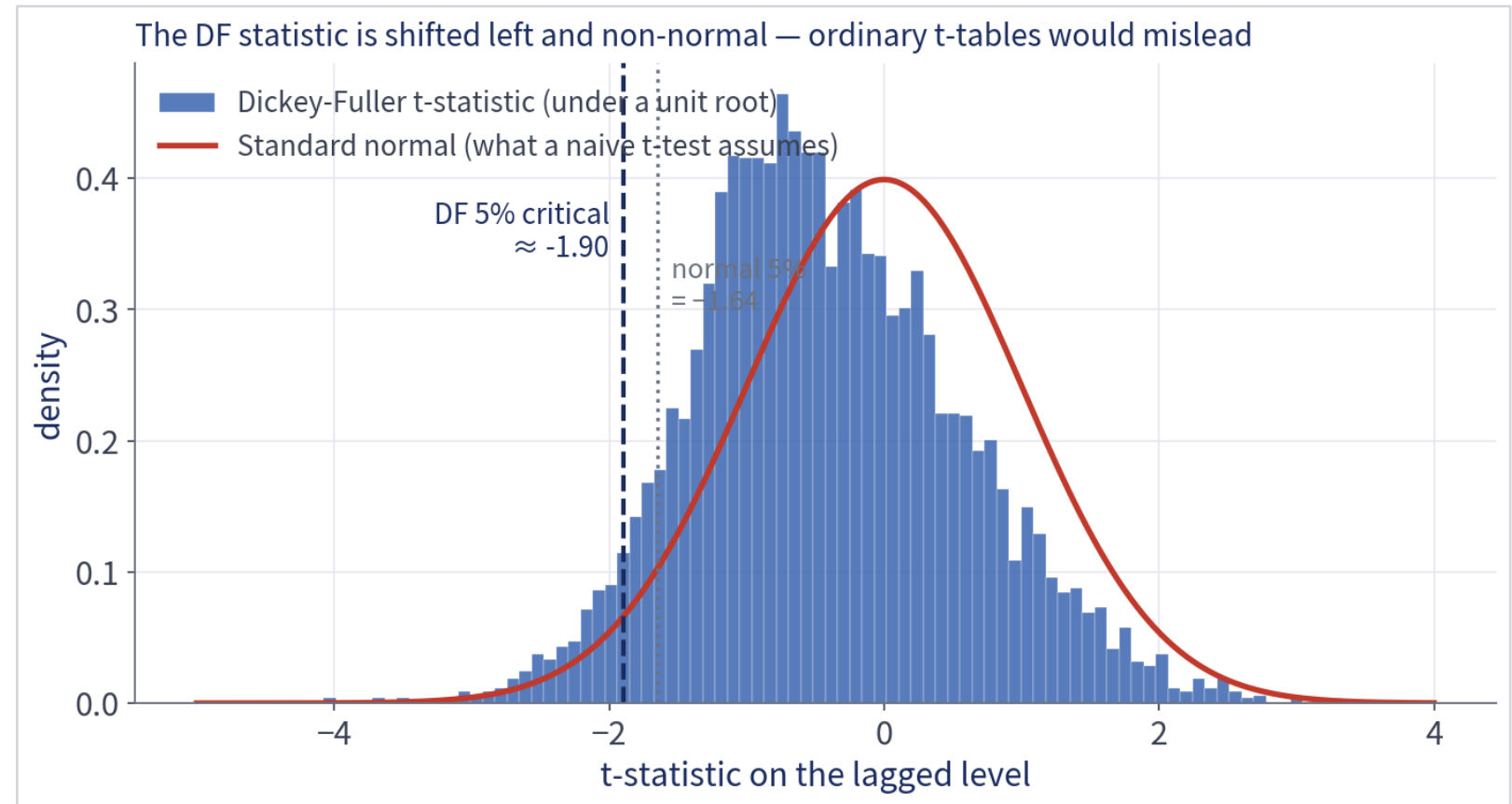
- Choose the lag length by an information criterion (AIC/BIC).
- Pick the specification (none/drift/trend) to match the plot.
- It is the default unit-root test in every stats package.

Why the ADF Needs Its Own Distribution

Under a unit root, the test statistic is not normal — so ordinary t-tables would mislead you.

What to notice

- Blue: the true DF stat.
- Red: the normal.
- The DF is shifted left.
- Its 5% cutoff is ~ -2.9 .
- Normal's is only -1.64 .
- Use DF tables, not t.



Choosing the ADF Specification

Match the deterministic part to what you see

Specification	What it includes	Use when the series ...
None	just the lagged level	fluctuates around zero (e.g. returns)
Drift (constant)	a constant + lagged level	has a non-zero mean or level
Trend	constant + time trend + level	shows a clear deterministic trend

Getting this wrong quietly destroys the test's power — always eyeball the plot first, then choose.

The KPSS Test: The Null Is Flipped

A confirmatory test that assumes the opposite

$$\underbrace{H_0^{\text{ADF}} : \text{unit root}}_{\text{reject} \Rightarrow \text{stationary}}$$

$$\underbrace{H_0^{\text{KPSS}} : \text{stationary}}_{\text{reject} \Rightarrow \text{unit root}}$$

The key difference

- ADF's null is a unit root; KPSS's null is stationarity.
- So rejecting KPSS is evidence AGAINST stationarity.
- Reading it like ADF is the most common student error.

Why flip the null

- The two tests fail in different situations.
- Running both guards against each one's blind spots.
- Agreement between them is what gives confidence.

Phillips-Perron and the Test Family

More than one way to hunt a unit root

Phillips-Perron (PP)

- Same null as ADF (a unit root), different machinery.
- Corrects for autocorrelation non-parametrically, without adding lags.
- A useful robustness check alongside the ADF.

The broader toolkit

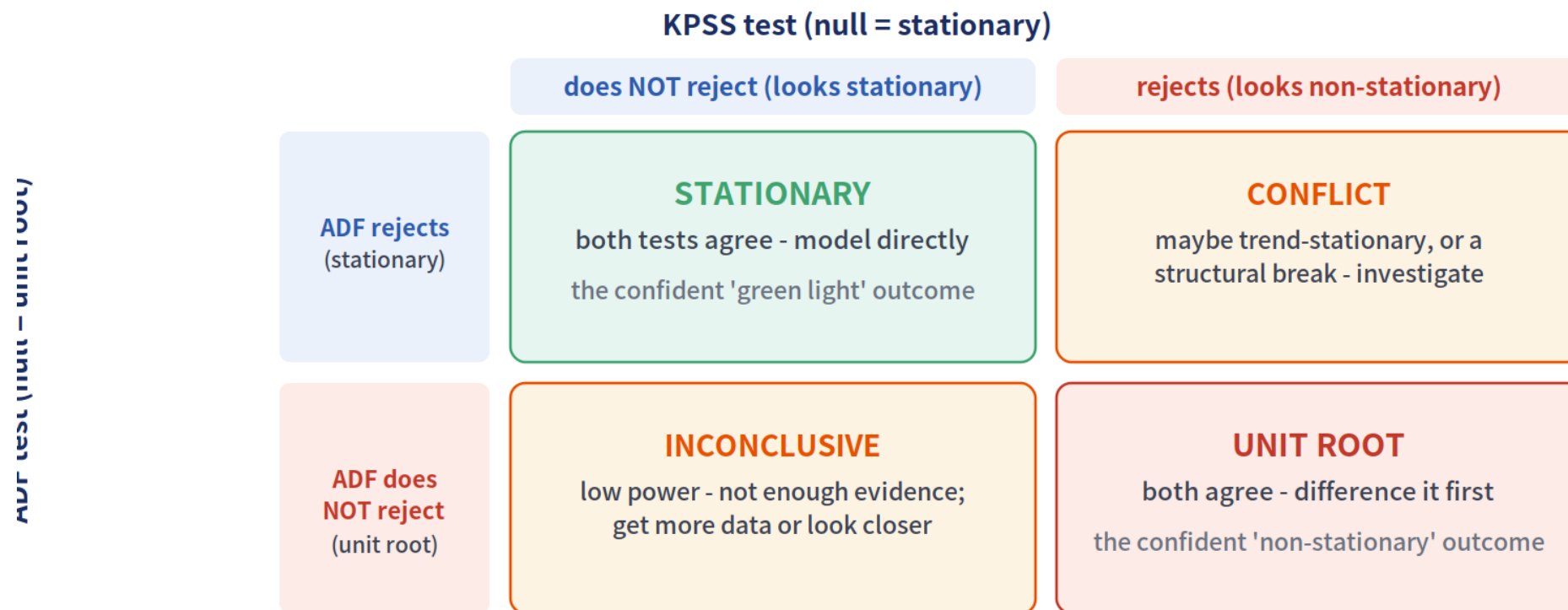
- ADF: parametric, adds lagged differences (the default).
- KPSS: opposite null, used for confirmation.
- DF-GLS: a more powerful variant when power is a concern.

ADF + KPSS: The Confirmatory 2×2

Because the two tests have opposite nulls, running both and reading the grid gives a far more confident verdict than either alone.

ADF + KPSS: The Confirmatory 2x2

The two tests have OPPOSITE nulls. Run both — agreement gives you a confident verdict.



Reading the Test Output

Statistics, critical values, and p-values

The mechanics

- Compare the ADF statistic to Dickey-Fuller critical values, not normal ones.
- A more negative statistic means stronger evidence against a unit root.
- Or simply read the reported p-value: small $\rightarrow$ reject the unit root.

Common mistakes

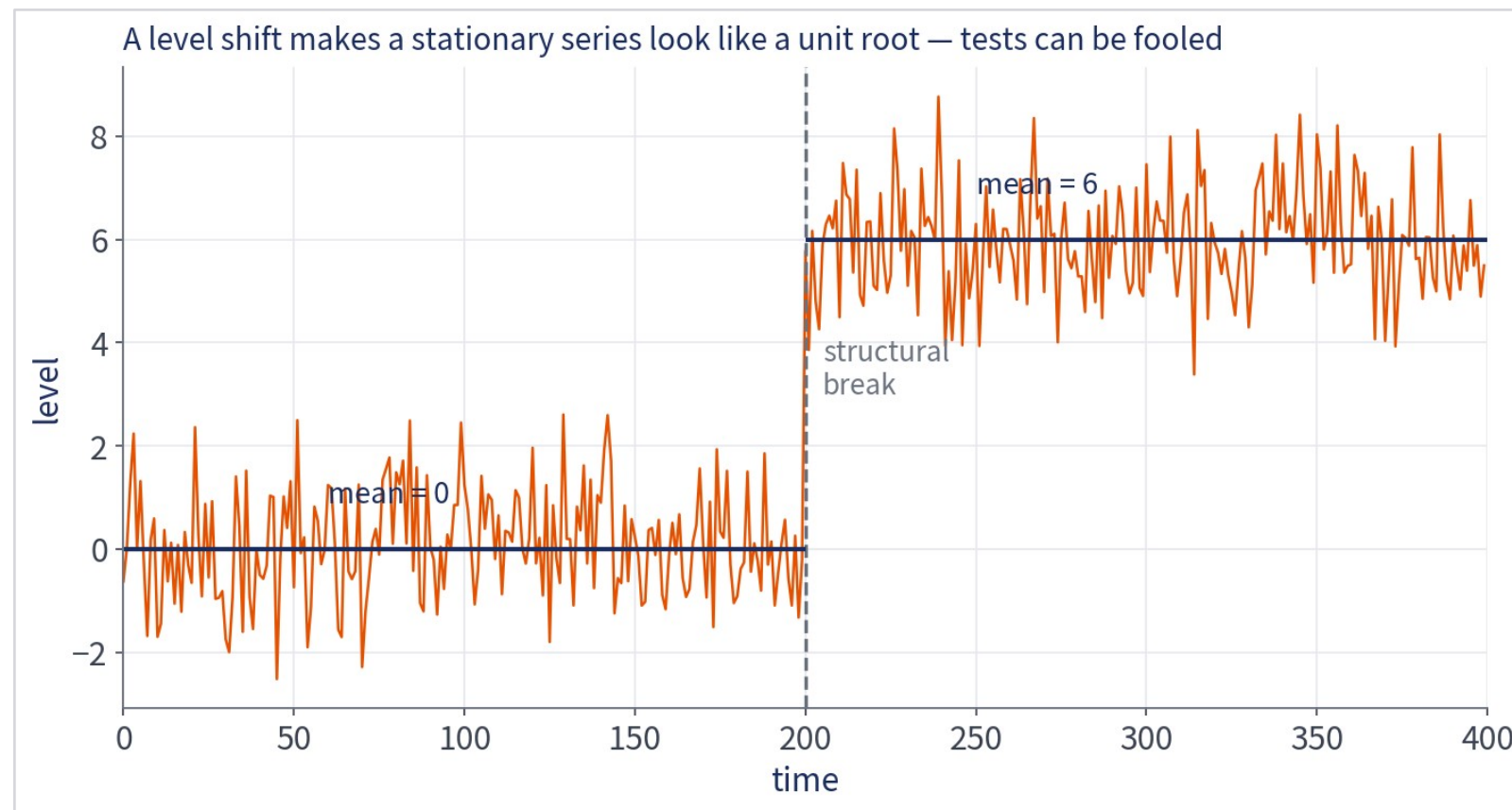
- Using ordinary t critical values (they are too lenient).
- Forgetting the specification changes the critical values.
- Reading KPSS like ADF — its rejection means the opposite.

When the Tests Struggle

Unit-root tests have low power and can be fooled by structural breaks.

Two blind spots

- Low power: near-unit
- roots often escape.
- Structural breaks:
 - a level shift mimics
 - a unit root here.
- Always plot first.



Making a Series Stationary

Differencing vs detrending — pick the right one

The two non-stationarities need two different cures. Applying the wrong one leaves the series non-stationary or destroys real structure.

Difference (for a unit root)

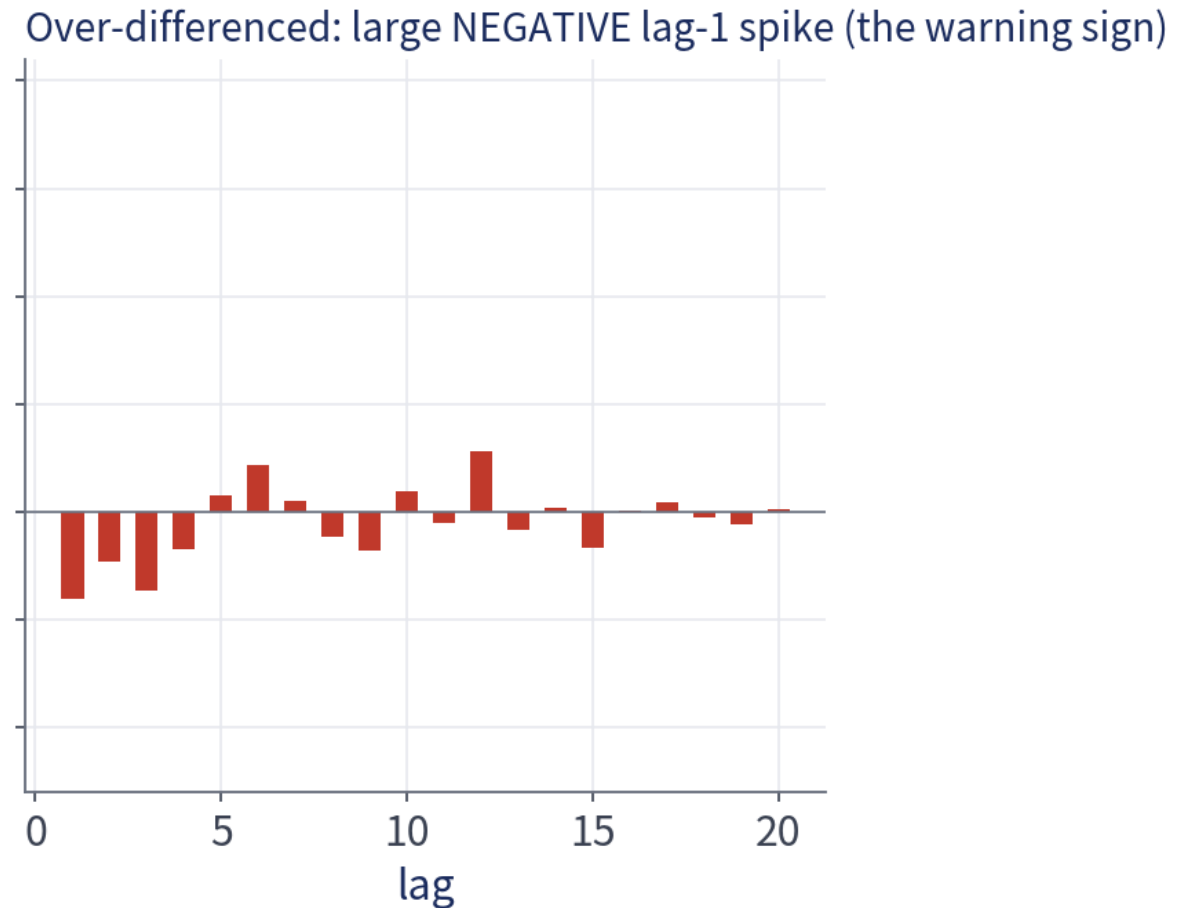
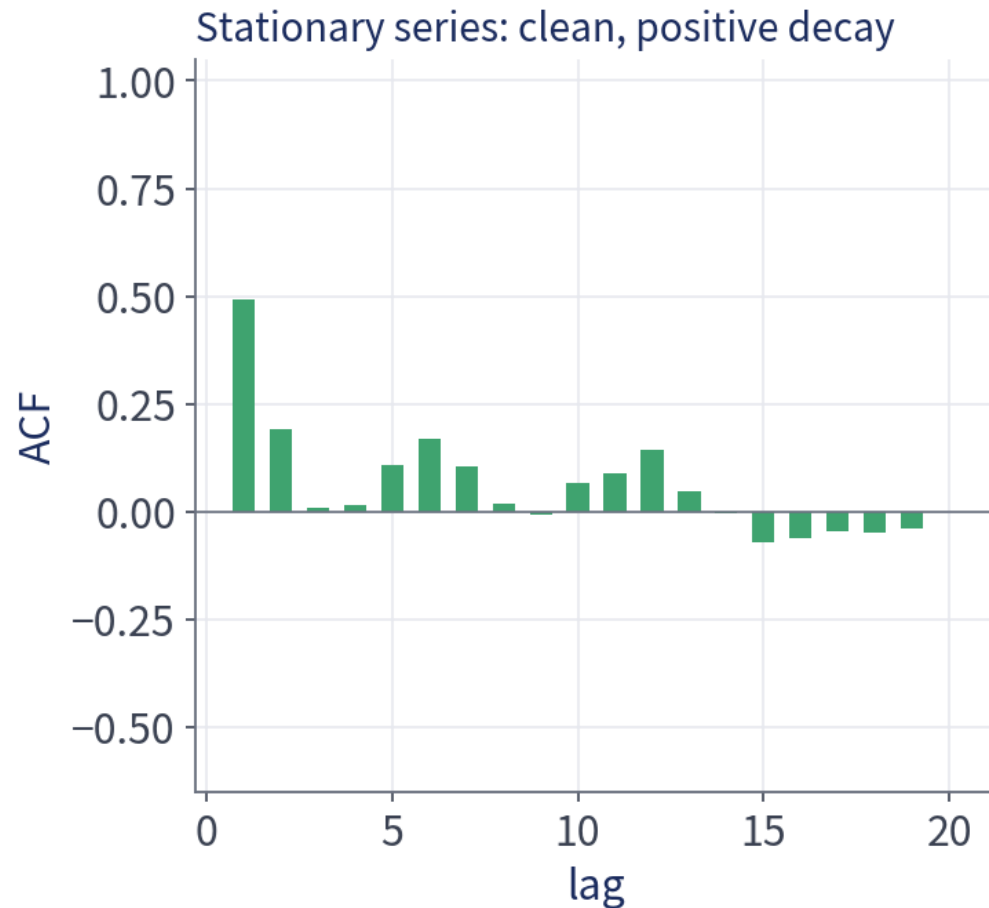
- Model ΔX_t instead of X_t — e.g. returns from prices.
- Removes a stochastic trend; confirmed by ADF/KPSS.
- The standard fix for $I(1)$ financial series.

Detrend (for a fixed trend)

- Subtract a fitted deterministic trend $a + b \cdot t$.
- Use only when the trend is genuinely deterministic.
- Differencing a trend-stationary series over-differences it →

The Over-Differencing Warning

Differencing an already-stationary series injects a large **NEGATIVE** lag-1 autocorrelation — the classic sign you have differenced one time too many.



Integration Order in Practice

Typical orders for common financial series

Series	Typical order	Cure to make it stationary
Asset price level	$I(1)$	difference $\rightarrow$ simple/log returns
Log returns	$I(0)$	already stationary — model directly
Interest-rate level	often $I(1)$	difference (test to confirm)
Inflation rate	$I(0)$ or $I(1)$	test; sometimes difference
Log GDP	$I(1)$	difference $\rightarrow$ growth rate

The Unit-Root Testing Workflow

A repeatable five-step routine

1 • Plot

eyeball

Look for trend, drift, or a break; pick the ADF spec.

2 • ADF

unit-root null

Run ADF; a small p-value rejects the unit root.

3 • KPSS

confirm

Run KPSS; check its verdict agrees with ADF.

4 • Difference

if needed

If a unit root, difference once and re-test.

5 • Re-check

no over-diff

Confirm stationary and watch for a negative lag-1 ACF.

Same routine, every series — this is what you will automate in the lab.

Building the Stationarity Diagnostic

A meta-prompt — you set the goal, Claude assembles the tests

GOAL Turn Part 2 into a reusable diagnostic that runs ADF + KPSS and recommends the cure — all from one prompt.

[INPUT] Meta-prompt · Claude Code

```
Using returns.csv (and the raw price level),  
write and run stationarity_test.py that, for  
any series I pass:  
1. plots it with its rolling mean and std;  
2. runs an ADF test (auto-selecting the spec  
   and lag length) and a KPSS test;  
3. prints both statistics, p-values, and a  
   combined verdict using the ADF+KPSS 2x2;  
4. if non-stationary, differences once and  
   re-tests, warning me if it over-differences.  
Run it on the price LEVEL and on returns.
```

How to work with it

- Expect the price level to be $I(1)$, returns $I(0)$.
- Read the 2×2 verdict, not just one p-value.
- Feed it a trending series — does it pick 'trend'?
- Keep it; Part 3 builds spurious regression on top.

Deliverable: stationarity_test.py — your reusable unit-root diagnostic.

Three Questions Before You Move On

If you can answer these, Part 2 has landed

Q1

ADF p-value = 0.84 and KPSS p-value = 0.01 on the same series. Verdict?

**Unit root —
difference**

Q2

You difference a stationary series and see a big negative lag-1 ACF. What happened?

Over-differenced

Q3

Why can't you judge the ADF statistic with ordinary t-tables?

Non-normal DF dist

All three come straight from the ADF/KPSS logic and the DF distribution.

Three Pitfalls to Avoid

Where unit-root testing goes wrong

Common mistakes

- "A high ADF p-value proves the series is non-stationary."
- "One test is enough."
- "When in doubt, just difference."

The reality

- Failing to reject is not proof — tests have low power; confirm with KPSS.
- Run ADF and KPSS together; disagreement is itself information.
- Over-differencing injects a negative lag-1 ACF and hurts forecasts.

Three Things to Carry Forward

Part 2 in one screen

1

A unit root is $\phi = 1$ — the boundary into non-stationarity

At that knife's edge, shocks are permanent and variance is unbounded. The random walk is the canonical example.

2

Test with ADF and KPSS together

They have opposite nulls and different blind spots. Agreement gives a confident verdict; the DF distribution — not the normal — sets the critical values.

3

Difference a unit root, detrend a fixed trend

Match the cure to the diagnosis, difference only once, and watch for the negative lag-1 ACF that signals over-differencing.

Next — Part 3: the spurious-regression trap in full, and its cure — cointegration.

★ WEEK 2 · PART 2 · PRACTICE

Unit-Root Tests, Worked by Hand

Run the ADF, Then Confirm It With KPSS

The Dickey-Fuller statistic, its special cutoff, and the 2x2 grid

Part 2's decision tool, with the arithmetic shown. We run the ADF regression and compute its Dickey-Fuller statistic, compare it to the special critical value, then confirm with KPSS's opposite null - and use the two together to fix a series' order of integration.

2P

PART 2 · APPLICATIONS

Random Walk, or Mean Reverter?

Case A: run the ADF test on a spread - compute the Dickey-Fuller statistic and compare it to the special cutoff - to decide random walk versus stationary. Case B: confirm with the ADF-plus-KPSS 2x2 grid and pin down the order of integration. Then a Claude Code lab.

What Part 2 Told Us

Now we apply it - with numbers

1

A unit root is an AR coefficient of one

When ϕ equals one, the series is a random walk - non-stationary. The ADF test asks whether it does.

2

The ADF uses special critical values

Under a unit root, the statistic follows the Dickey-Fuller distribution, not the normal - so the cutoff is more negative.

3

ADF and KPSS confirm each other

They have opposite nulls, so running both and reading the 2x2 grid gives a confident, cross-checked verdict.

Everything here is a calculator exercise: one Dickey-Fuller statistic, one confirmatory grid.

How These Cases Work

Data, then arithmetic, then meaning

Each case has three steps

- Start from a series and its ADF regression.
- Compute the statistic, or read the grid.
- Then decide stationarity.

What you should watch

- How the DF statistic beats a special cutoff.
- Why the cutoff is more negative than normal.
- How KPSS confirms the ADF.

The ADF Test

Random walk, or tradeable spread?

Before trading a spread, we must know it mean-reverts rather than wandering. The ADF regression tests exactly that: it estimates the coefficient on the lagged level and asks whether it's zero - a unit root. We compute the Dickey-Fuller statistic for a candidate spread, compare it to the special cutoff and reach a verdict

The setup

- A candidate pairs-trade spread.
- The ADF coefficient estimate is -0.18.
- Its standard error is 0.05.

What we'll do by hand

- Form the Dickey-Fuller statistic.
- Compare it to the special cutoff.
- Decide random walk or stationary.

The ADF Regression

Testing the lagged level

$$\Delta y_t = \alpha + \gamma y_{t-1} + \sum \delta_i \Delta y_{t-i} + \varepsilon_t; \quad H_0 : \gamma = 0 \text{ (unit root, since } \gamma = \phi - 1)$$

The setup

- Regress the change on the lagged level.
- The coefficient gamma is phi minus one.
- Extra lagged changes absorb short-run dynamics.

The null

- Gamma equals zero means phi equals one.
- That's a unit root - a random walk.
- Rejecting it means stationary.

The Dickey-Fuller Statistic

Coefficient over its standard error

$$\text{spread: } t = \frac{\hat{\gamma}}{\text{se}} = \frac{-0.18}{0.05} = -\mathbf{3.60} \quad (\hat{\phi} = 1 + \hat{\gamma} = 0.82, \text{ mean-reverting})$$

The calculation

- Minus 0.18 over 0.05 is minus 3.60.
- The implied phi is 0.82.
- Well below one - it mean-reverts.

What it suggests

- A strongly negative statistic.
- The lagged level pulls the change back.
- But we must use the right cutoff.

The Verdict

Against the special cutoff

$$-3.60 < \underbrace{-2.86}_{\text{DF 5\%}} < \underbrace{-1.65}_{\text{normal 5\%}} \Rightarrow \text{reject unit root} \Rightarrow \text{stationary (tradeable)}$$

Why special

- Under a unit root, the statistic isn't normal.
- It follows the Dickey-Fuller distribution.
- Shifted left, so the cutoff is more negative.

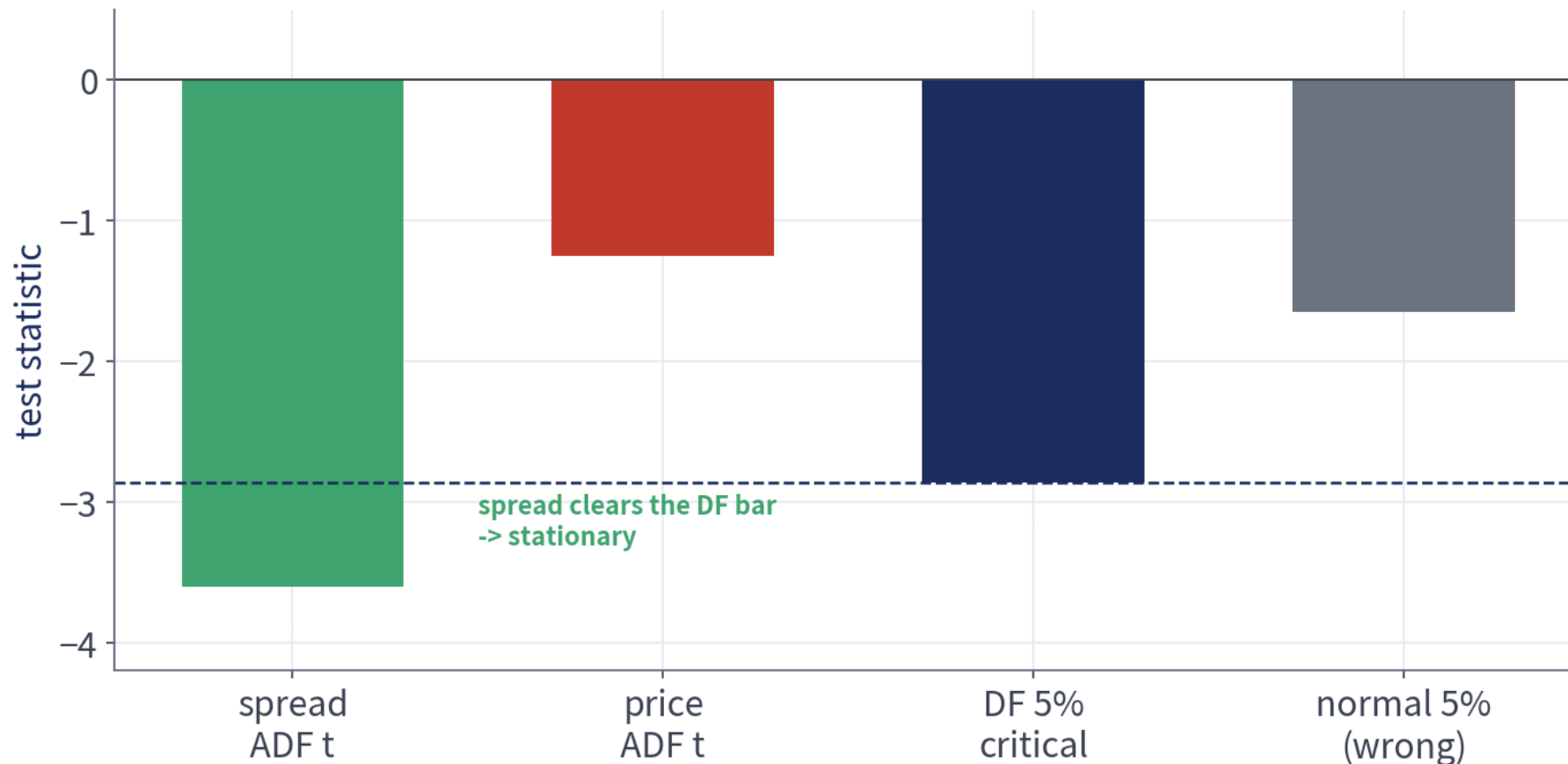
The decision

- At -3.60, it clears -2.86.
- Reject the unit root - the spread is stationary.
- The pair is tradeable.

Clearing the Right Bar

The spread's ADF statistic of -3.60 sits below the Dickey-Fuller cutoff of -2.86, so it's stationary. A price would give roughly -1.25, failing to clear it - a random walk. Note the DF cutoff is more negative than the normal -1.65: using the normal would wrongly declare random walks stationary

Case A: only the spread's ADF stat clears the Dickey-Fuller cutoff



Why the Diagnosis Matters

Spurious regression waits

The stakes

- Regressing two random walks looks significant.
- But the relationship is spurious.
- Non-stationarity fabricates false structure.

The safeguard

- Test for a unit root before regressing.
- Trade only spreads that reject it.
- The right cutoff prevents false confidence.

Confirm, and Count the Differences

ADF plus KPSS, and the order

One test can mislead, so we confirm. KPSS has the opposite null - it assumes stationarity - so pairing it with the ADF gives a cross-check. We read the 2x2 grid for a price and its returns, and in doing so pin down the order of integration: how many differences make the series stationary

The two tests

- ADF null: there is a unit root.
- KPSS null: the series is stationary.
- Opposite nulls, so they cross-check.

What we'll do by hand

- Read the grid for the price level.
- Then for its returns.
- Confirm the price is integrated of order one.

The Confirmatory Grid

Opposite nulls, read together

ADF H_0 : unit root $\perp$ KPSS H_0 : stationary (opposite nulls $\Rightarrow$ run both, read the grid)

Why two tests

- A single test can fail to reject weakly.
- The opposite null gives independent evidence.
- Agreement is a confident call.

Reading the corners

- ADF rejects and KPSS doesn't: stationary.
- ADF doesn't and KPSS rejects: unit root.
- Disagreement means inconclusive.

The Grid, Filled In

Price level versus returns

series	ADF	KPSS	verdict
price level	t=-1.25 (fail)	0.85 (reject)	unit root
returns	t=-8.50 (reject)	0.12 (fail)	stationary

For the price level, the ADF fails to reject and KPSS rejects stationarity - both point to unit root. For the returns, the ADF rejects and KPSS fails to reject - both point to stationarity. The two tests agree in each row, giving confident verdicts.

The Order of Integration

How many differences?

price level: ADF fails, KPSS rejects $\rightarrow$ difference $\rightarrow$ ADF rejects, KPSS fails $\Rightarrow$ price is $I(1)$

The logic

- The level is non-stationary - a unit root.
- One difference makes it stationary.
- So the price is integrated of order one.

In finance

- Prices are $I(1)$; returns are $I(0)$.
- That's why we model returns, not prices.
- The order tells you how much to difference.

The 2x2 Verdict

Where agreement is confident

Reading the grid

- Columns: does ADF reject?
- Rows: does KPSS reject?

ADF rejects (stationary) ADF fails (unit root)

Case B: opposite nulls -> the agreeing corners are confident calls

KPSS fails (stationary)	STATIONARY (confident)	inconclusive
KPSS rejects (unit root)	inconclusive	UNIT ROOT (confident)

price level -> UNIT ROOT cell; returns -> STATIONARY cell

confident;

- the off-diagonal is inconclusive.

Confirm Before You Difference

And difference the right amount

Why confirm

- A weak ADF result alone can mislead.
- KPSS's opposite null catches it.
- Agreement is what you act on.

Why the order matters

- Under-differencing leaves a unit root.
- Over-differencing injects noise.
- The order tells you exactly how much.

The Unit-Root Workflow

Test, confirm, difference

The routine is disciplined. Run the ADF to test for a unit root, comparing against its special Dickey-Fuller cutoff, not the normal. Confirm with KPSS, whose opposite null cross-checks the result - trust only the agreeing corners of the grid. Then difference just enough to reach stationarity, fixing the order of integration. That

The test (Case A)

- Spread ADF -3.60 clears -2.86.
- Stationary - tradeable.
- Price fails - a random walk.

The confirmation (Case B)

- Level: both tests say unit root.
- Returns: both say stationary.
- So the price is $I(1)$.

Reproduce the ADF and the Grid

Direct Claude Code to check your arithmetic

GOAL Run the ADF test, compare to the DF cutoff, and confirm with KPSS.

[INPUT] Prompt · Claude Code

```
Write and run part2_practice.py that:  
CASE A - ADF:  
  * simulate a stationary spread and a random-walk  
    price; run adfuller on each;  
  * print the statistic and compare to -2.86.  
CASE B - confirm and order:  
  * run adfuller and kpss on a price level and its  
    returns; print both statistics;  
  * report the 2x2 verdict and the order of integration.  
Then plot the ADF statistics and the 2x2 grid.
```

Check against the slides

- spread ADF $-3.60 < -2.86$.
- price ADF -1.25 (fail).
- level: ADF fail, KPSS reject.
- returns: ADF reject, KPSS fail $\rightarrow I(1)$.

Deliverable: part2_practice.py whose printed numbers match the ADF verdicts and the grid.

Three Questions on the Cases

If you can answer these, the practice has landed

Q1 What does the ADF null hypothesis say?

That the series has a unit root - it's non-stationary

Q2 Why use special critical values?

Under a unit root the statistic isn't normally distributed

Q3 Why pair ADF with KPSS?

Opposite nulls give a confident, cross-checked verdict

All three come from the ADF regression, its special cutoff, and the confirmatory grid.

Three Pitfalls to Avoid

Where unit-root testing trips people

Common mistakes

- "Compare the ADF statistic to normal cutoffs."
- "Failing to reject proves stationarity."
- "One test settles it."

The reality

- Use Dickey-Fuller critical values - not the normal.
- Failing to reject is weak evidence, not proof.
- Confirm with KPSS; trust the agreeing corners.

Two Calculations to Remember

Part 2 applications in one screen

1

The ADF statistic clears a special cutoff

The spread's Dickey-Fuller statistic was minus 0.18 over 0.05, or -3.60, implying a ϕ of 0.82. It cleared the special cutoff of -2.86 - more negative than the normal -1.65 - so we reject the unit root: the spread is stationary and tradeable.

2

ADF and KPSS confirm the order

For the price level, the ADF failed to reject and KPSS rejected stationarity - both say unit root. For the returns, the ADF rejected and KPSS failed to reject - both say stationary. The agreement pins the price as integrated of order one.

3

Test, confirm, then difference

Run the ADF against its special cutoff, confirm with KPSS's opposite null, and difference just enough to reach stationarity. That order of integration is the foundation every later model rests on.

Next — Part 3 practice: cointegration, when two non-stationary series share a stationary combination.

★ WEEK 2 · PART 3 of 3

Spurious Regression & Cointegration

The Trap, and the Cure

Why unrelated series fake relationships — and how to find the real ones

Now we deliver the payoff of the 1974 episode: the exact mechanism of spurious regression, and cointegration — the elegant idea that separates a genuine long-run link from a statistical mirage.

3

PART 3 · COINTEGRATION

From the Trap to the Cure

The spurious-regression mechanism in full, then cointegration, the error-correction model, and the application that made both famous — pairs trading.

The Episode's Unfinished Business

We named the trap in Part 1 — now we open it up

1

Part 1 warned us; Part 2 armed us

We can now detect a unit root with ADF and KPSS. That is exactly the tool we need to understand — and defuse — spurious regression.

2

The trap: two $I(1)$ series can fake a relationship

Regressing one trending series on another unrelated one yields a high R^2 and a huge t-statistic that mean nothing at all.

3

The cure has a name: cointegration

Sometimes two $I(1)$ series really do move together. Cointegration is how we tell a real long-run link from a mirage — and trade it.

Part 3: first the mechanism of the mirage, then the mathematics of the real thing.

The Spurious Regression Trap, Formally

Why the residual is the tell

$$y_t = \alpha + \beta x_t + u_t, \quad x_t, y_t \sim I(1) \text{ independent} \Rightarrow u_t \sim I(1)$$

What goes wrong

- Regress two independent $I(1)$ series and the residual is itself $I(1)$.
- A non-stationary residual violates the regression's core assumption.
- So every standard error and t-statistic is invalid.

The diagnostic

- The giveaway is a non-stationary residual, not a low R^2 .
- In fact R^2 is often deceptively high.
- Always test the residual before you trust the regression.

Why R^2 and the t-Statistic Explode

The usual asymptotics simply fail

as $T \rightarrow \infty$: $|t_{\hat{\beta}}| \rightarrow \infty$, $R^2 \not\rightarrow 0$ (the t-test never settles)

The broken machinery

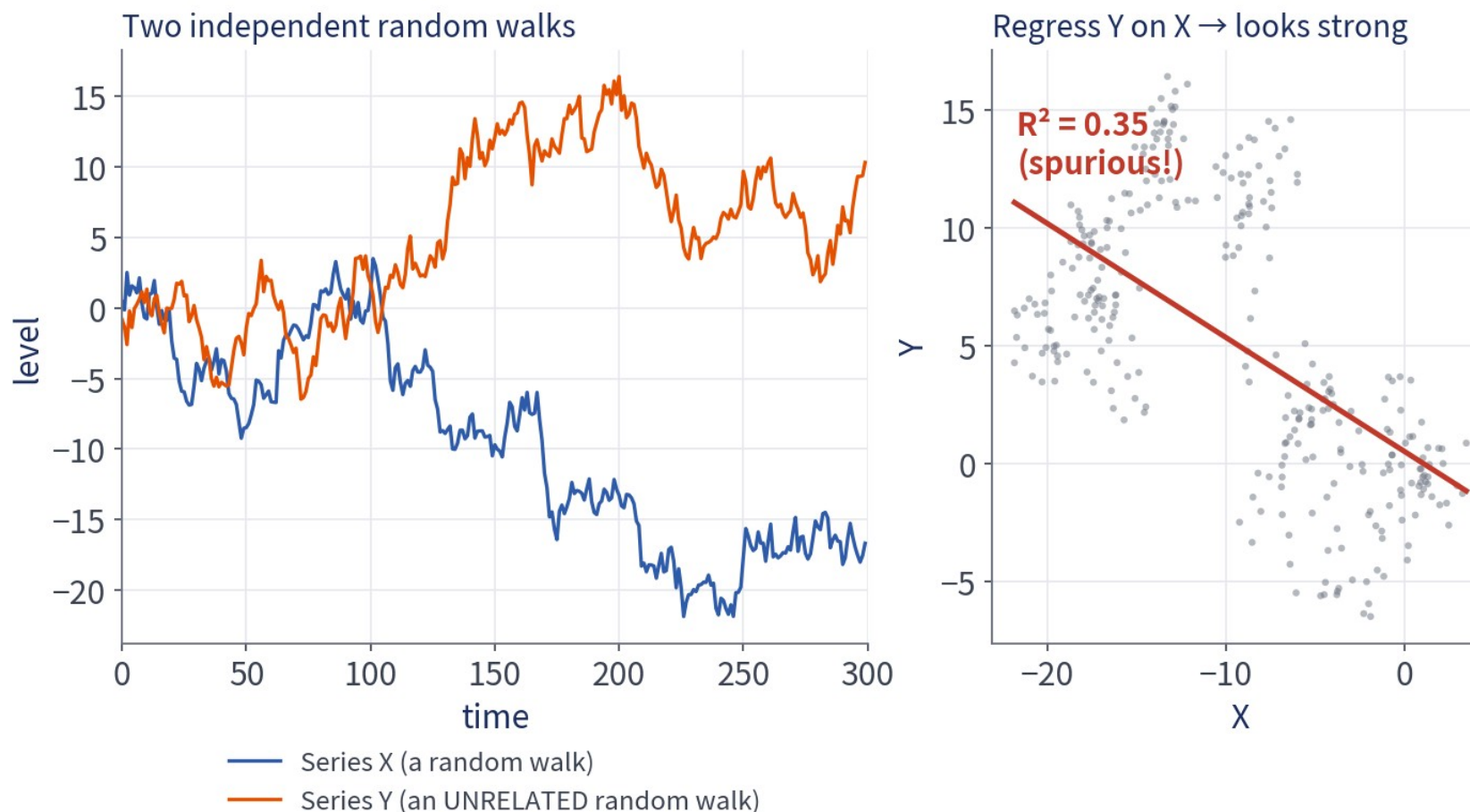
- For stationary data, the t-statistic settles to a fixed distribution.
- For two $I(1)$ series, it instead diverges as the sample grows.
- More data makes the illusion look STRONGER, not weaker.

The intuition

- Both series wander; by chance they drift in loose lockstep.
- The regression reads that shared wandering as 'correlation'.
- But there is no mechanism linking them — it is an artifact.

Two Unrelated Random Walks Fake a Relationship

Left: two independent random walks. Right: regress one on the other and you get a high R^2 and a confident-looking line — from series with no connection at all.

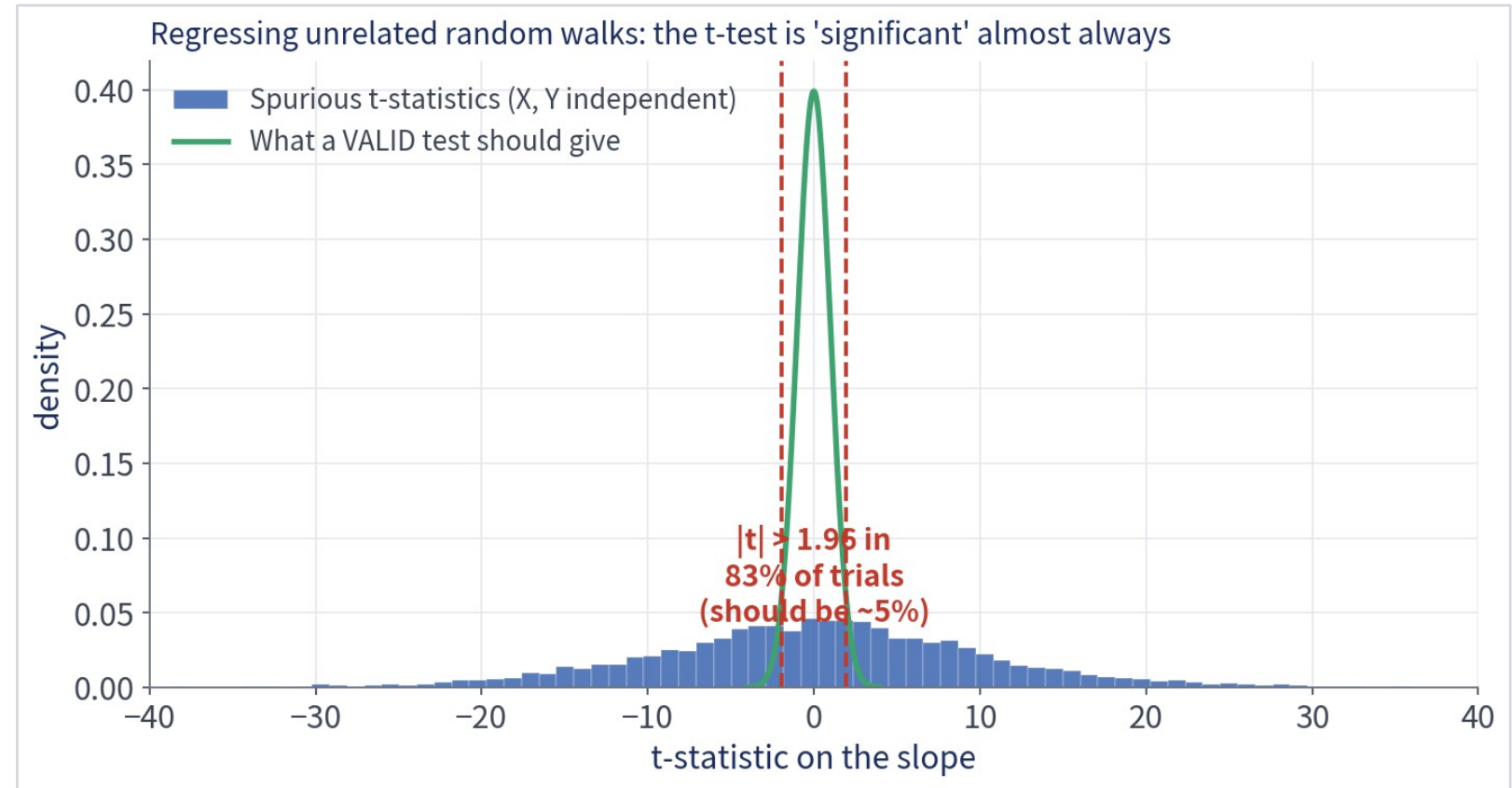


How Often the t-Test Lies

Repeat the experiment thousands of times: the t-test screams 'significant' almost every time.

What to notice

- Blue = spurious t's.
- They spread past ± 2 .
- Green = a valid test.
- $|t| > 2$ should be $\sim 5\%$.
- Here: the majority.



Granger and Newbold's Warning Sign

A quick check you can run on any regression

Warning sign: $R^2 > DW$ (Durbin–Watson) $\Rightarrow$ suspect spurious

How to read it

- The Durbin-Watson statistic detects residual autocorrelation.
- A very low DW alongside a high R^2 is a red flag.
- R^2 exceeding DW is the classic spurious-regression signature.

What to do next

- Do not trust the coefficients — investigate the residual.
- Test whether the residual is stationary ($I(0)$) or not.
- If it wanders, you are looking at a spurious regression.

How to Avoid Spurious Regression

Four habits that keep you honest

Do this	Instead of	Why it matters
Test each series for $I(1)$ first	regressing raw levels blindly	levels may share only a trend
Check cointegration before using levels	trusting a high R^2	R^2 is meaningless for $I(1)$ levels
Model differences or an ECM	modeling raw levels	differences/spread are stationary
Inspect the residual for $I(0)$	reading t-stats at face value	non-stationary residuals void inference

The single unifying habit: never regress two levels without checking what their residual does.

The Naive Fix: Regress Differences

Correct, but it comes at a cost

Differencing both series makes them stationary, so the regression becomes valid again. This is the safe, standard fix — but it quietly discards something valuable.

Why it works

- Differences of $I(1)$ series are $I(0)$ — stationary.
- On stationary data, t-tests and R^2 behave normally.
- For most purposes, model returns, not price levels.

What it loses

- Differencing removes the LONG-RUN relationship between levels.
- You can no longer say 'these two prices move together'.
- Sometimes that long-run link is exactly what you want →

But Differencing Throws Away the Long Run

When the levels genuinely belong together

Two prices can wander individually yet stay tethered — an arbitrage force pulls them back whenever they drift apart. Differencing erases that tether entirely.

The everyday example

- Two share classes of the same company.
- A stock and its close industry peer.
- Spot and futures on the same asset.

What we want instead

- A method that keeps the levels but stays valid.
- One that captures the tether, not just the wiggles.
- That method is cointegration.

Cointegration: A Real Long-Run Link

Two wanderers, one stationary combination

Two series are cointegrated if each is individually non-stationary, yet a particular linear combination of them IS stationary. That combination is the tether — and it mean-reverts.

The picture

- Each series alone is $I(1)$ — it wanders with no fixed mean.
- But their spread is $I(0)$ — it reverts to a stable level.
- The spread is what you actually model and trade.

Why it is deep

- It distinguishes a genuine equilibrium from a coincidence.
- Granger and Engle won the 2003 Nobel for formalizing it.
- It rescues the levels that differencing would have thrown away.

Cointegration, Formally

One equation captures the whole idea

$$X_t, Y_t \sim I(1), \quad \exists \beta : \quad Z_t = Y_t - \beta X_t \sim I(0) \quad (\text{cointegration})$$

Reading it

- X and Y are each $I(1)$ — non-stationary on their own.
- There exists a β making $Z = Y - \beta X$ stationary, $I(0)$.
- β is the cointegrating coefficient; Z is the spread.

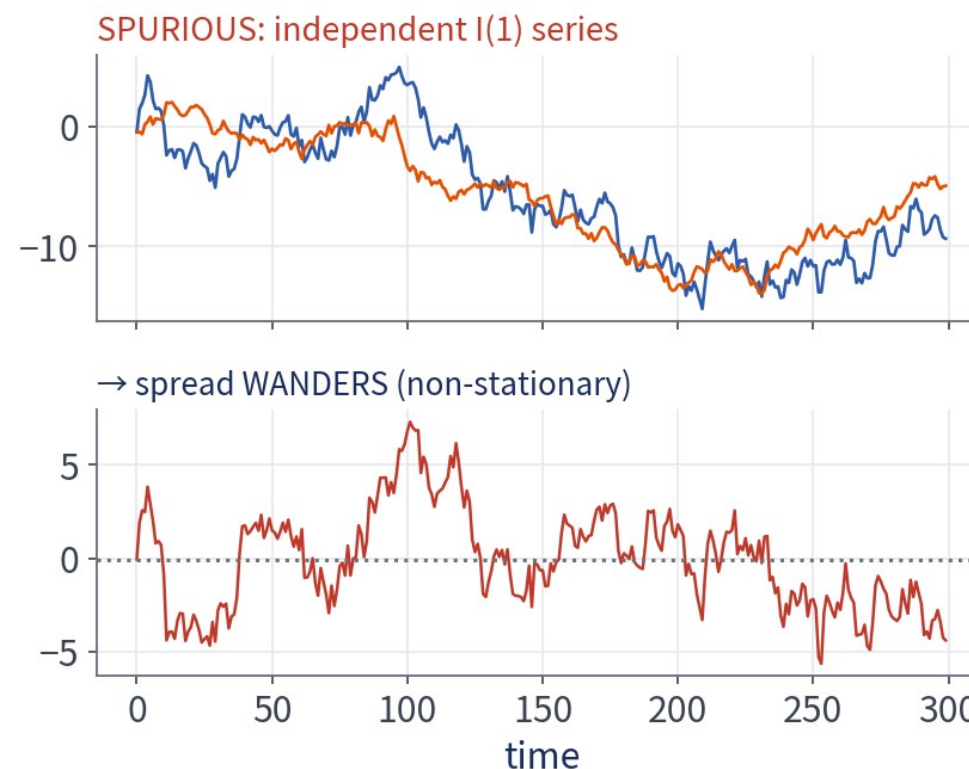
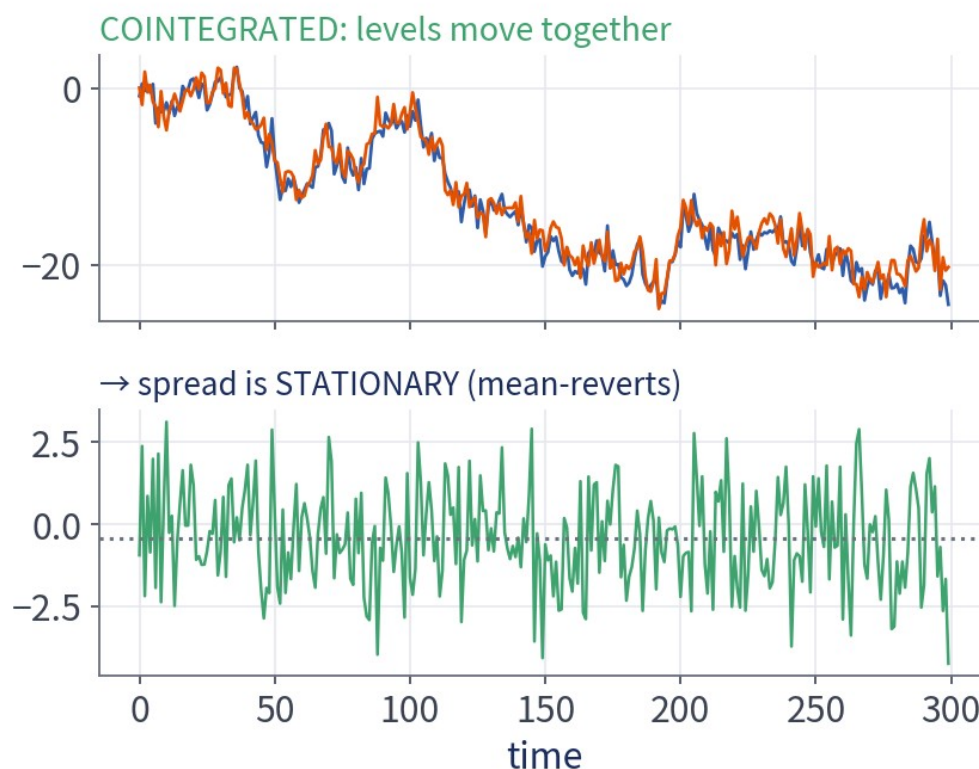
What it guarantees

- The spread has a fixed mean it keeps returning to.
- Deviations are temporary and self-correcting.
- So the levels carry real, usable long-run information.

Cointegrated vs Spurious — Look at the Spread

The single distinguishing test: if the spread between the levels is stationary (left), they are cointegrated; if it wanders (right), the regression is spurious.

Cointegration is what separates a real long-run link from a spurious one



The Engle-Granger Two-Step Test

Test cointegration with tools you already have

$$Y_t = \alpha + \beta X_t + Z_t$$

step 1: regress levels

ADF test on $\hat{Z}_t$

step 2: is it $I(0)$?

Step 1: regress the levels

- Regress Y on X and keep the residual, the estimated spread.
- This $\hat{\beta}$ is your candidate cointegrating coefficient.
- Ignore its t-statistic here — that is not the test.

Step 2: test the residual

- Run an ADF-type test on the residual.
- If the residual is $I(0)$, the series ARE cointegrated.
- If it is $I(1)$, the regression was spurious after all.

The Error Correction Model (ECM)

How the spread pulls the series back

$$\Delta Y_t = c + \gamma \underbrace{Z_{t-1}}_{\text{last error}} + \sum_j \phi_j \Delta X_{t-j} + \varepsilon_t, \quad \gamma < 0$$

The two forces

- Short-run: changes in X move Y period to period.
- Long-run: the last spread Z_{t-1} pulls Y back toward equilibrium.
- $\gamma < 0$ is the speed of that correction.

Why it is elegant

- It uses the levels (via the spread) AND stays valid.
- It unifies short-run dynamics with the long-run tether.
- It is the natural model once cointegration is confirmed.

The Johansen Test

When more than two series share equilibria

Engle-Granger handles a single pair. The Johansen test generalizes to many series at once and can find MORE than one cointegrating relationship among them.

What it adds

- A system approach for several $I(1)$ series together.
- It counts how many independent equilibria exist.
- The standard tool for baskets, curves, and multi-asset spreads.

When you need it

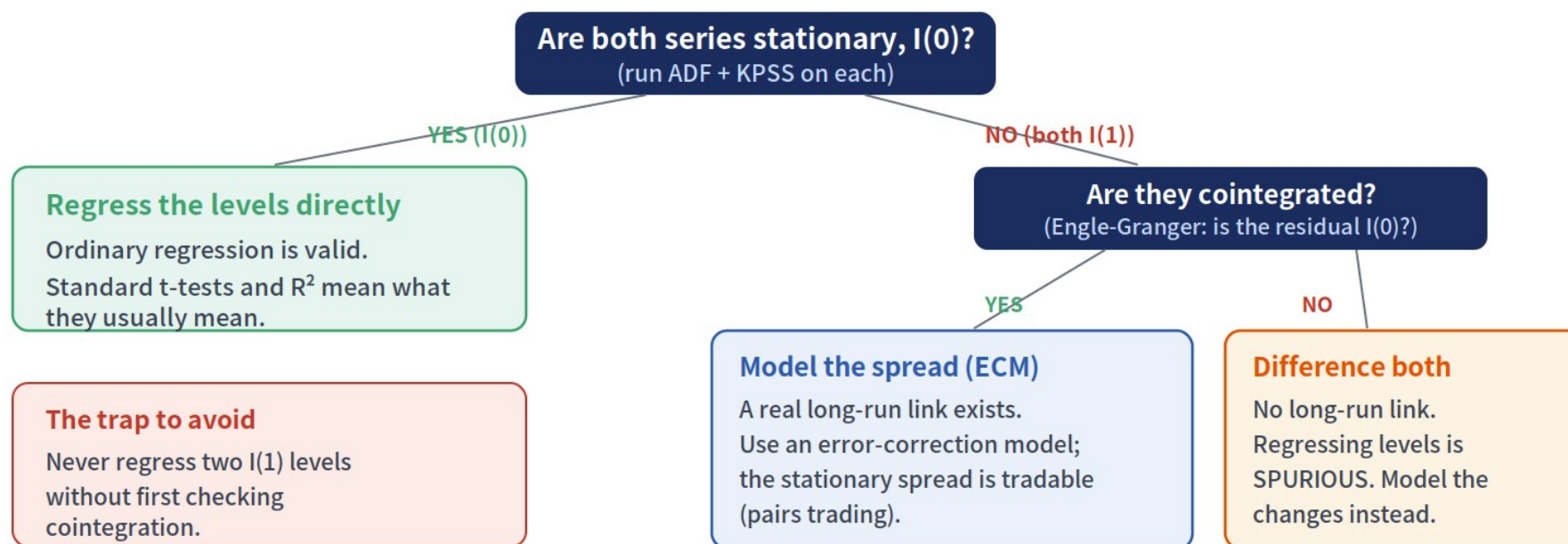
- Yield curves — many maturities tied together.
- A basket of related stocks or commodities.
- Any time a single pair is too narrow a lens.

Two Series, One Decision

The whole of Week 2 as a single flow: test for $I(1)$, then for cointegration, and let the answer choose your model — levels, the spread, or differences.

Two Series, One Decision: How to Regress Them Safely

The whole of Week 2, as a single flow. Follow it before you ever regress two levels.



Pairs Trading: Cointegration at Work

The strategy the theory makes possible

If two assets are cointegrated, their spread mean-reverts. Sell the spread when it is stretched wide, buy it when it is compressed, and profit as it returns to its mean.

The recipe

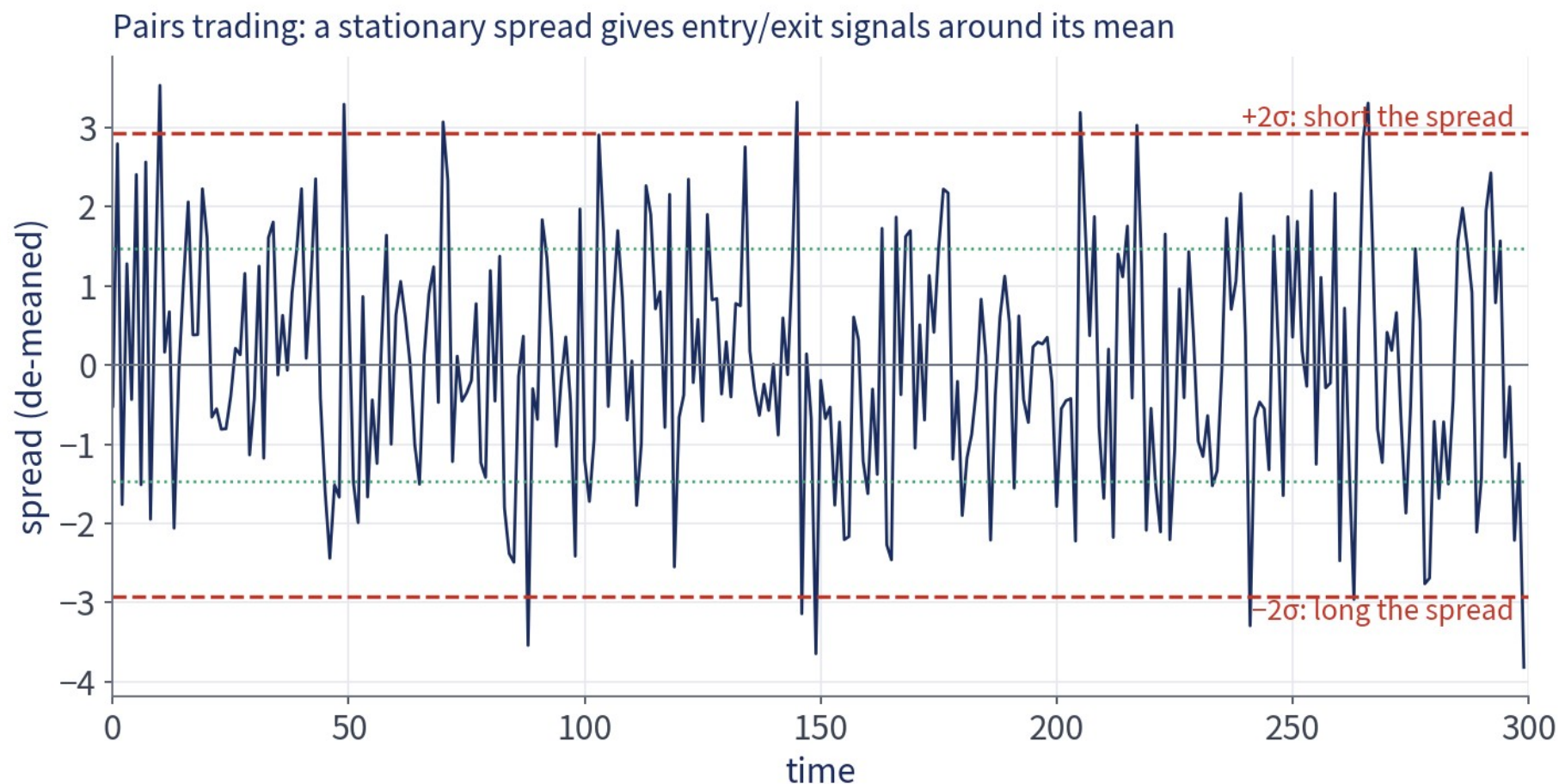
- Find a cointegrated pair; estimate the spread and its mean.
- Enter when the spread deviates far (e.g. ± 2 standard deviations).
- Exit as it reverts back toward the mean.

Why it can work

- The trade is market-neutral — it bets on the tether, not direction.
- Cointegration is the statistical license for 'reversion to a mean'.
- Recall Part 1: reversion needs a stationary series — here, the spread.

The Tradable Spread

A stationary spread oscillates around its mean. The dashed bands mark where it is 'stretched' — the entry signals for a mean-reversion (pairs) trade.



Cointegration Can Break

The tether is not guaranteed to hold

Where it fails

- A structural break — a merger, a regime change — can snap the link.
- A relationship estimated in-sample may not persist out-of-sample.
- Transaction costs can erase a small, slow-reverting edge.

How to stay safe

- Re-test the relationship regularly; do not assume it is permanent.
- Validate out-of-sample before committing capital.
- Size positions for the day the spread does NOT revert.

The Two-Series Workflow

From raw levels to a valid model

1 • Test $I(1)$

each series

Confirm both are individually non-stationary (Part 2).

2 • Regress

get the spread

Regress the levels; keep the residual as the spread.

3 • Test spread

cointegration

ADF-test the residual: is it $I(0)$?

4 • Model

ECM or diff

If cointegrated, build an ECM; if not, difference.

5 • Monitor

stay honest

Re-test over time; the tether can break.

This flow turns the spurious-regression trap into a disciplined, testable routine.

Reproducing 1974, then Finding a Real Pair

Direct Claude Code — see the trap and the cure yourself

GOAL Reproduce the spurious-regression result, then test two real assets for cointegration and build the spread.

[INPUT] Prompt · Claude Code

Write and run `cointegration_lab.py` that:

1. simulates 1000 pairs of INDEPENDENT random walks, regresses each pair, and reports how often $|t| > 1.96$ (the spurious rate);
 2. downloads two related tickers (e.g. KO and PEP), tests each for $I(1)$, then runs the Engle-Granger cointegration test on them;
 3. if cointegrated, plots the spread with its mean and $\pm 2\sigma$ bands and marks entry signals.
- Explain, in plain English, what each result means.

How to work with it

- Expect the spurious rate near 70–90%, not 5%.
- Check the residual's ADF p-value for the pair.
- Try an UNrelated pair — cointegration should fail.
- This ties Parts 1–3 into one workflow.

Deliverable: `cointegration_lab.py` + a spread chart for your chosen pair.

Three Questions Before You Move On

If you can answer these, Part 3 has landed

Q1

You regress two $I(1)$ prices and get $R^2 = 0.9$, huge t-stat. Believe it?

No — test residual

Q2

Both series are $I(1)$ and their spread is $I(0)$. What can you build?

An ECM / pairs trade

Q3

Why is differencing a valid but sometimes wasteful fix?

Loses the long run

All three hinge on one question: what is the residual doing?

Three Pitfalls to Avoid

Where the cure gets misapplied

Common mistakes

- "A high R^2 between two prices proves a real relationship."
- "Cointegration, once found, lasts forever."
- "The Engle-Granger t-stat in step 1 is the test."

The reality

- R^2 on $I(1)$ levels is meaningless — only the residual's order matters.
- Relationships break; re-test and validate out-of-sample.
- The test is the ADF on the residual (step 2), not step 1's t-stat.

Week 2 in Three Ideas

Pulling the whole week together

1

Stationarity is the gatekeeper

Part 1: model only stable statistics. Weak stationarity, the ACF, and the AR(1) lens tell a mean-reverting series from a wandering one.

2

Unit roots are detected and cured by testing

Part 2: ADF and KPSS together decide; a unit root is differenced, a deterministic trend is detrended — never the reverse.

3

Cointegration separates real links from mirages

Part 3: unrelated $I(1)$ series fake relationships; a stationary spread reveals the genuine ones — the basis of the error-correction model and pairs trading.

Next — Week 3: ARMA Models, building forecasts on the stationary series you now know how to make.

★ WEEK 2 · PART 3 · PRACTICE

Cointegration, Worked by Hand

Catch a Spurious Regression, Then Find the Real Long-Run Link

The Granger-Newbold warning, and the Engle-Granger residual test

Part 2's finale, with the arithmetic shown. We expose a spurious regression - a high R-squared betrayed by a near-zero Durbin-Watson - then use the residual to tell it apart from genuine cointegration, where a stationary combination reveals a real, tradeable equilibrium.

3P

PART 3 · APPLICATIONS

Fake Link, or Real Equilibrium?

Case A: two independent random walks produce a high R-squared and a huge t-statistic - a spurious regression the Durbin-Watson exposes. Case B: when the link is genuine, the residual is stationary - cointegration - and the Engle-Granger test confirms it. The residual is the tell. Then a Claude Code lab.

What Part 3 Told Us

Now we apply it - with numbers

1

Regressing random walks is a trap

Two unrelated non-stationary series can show a high R-squared and a huge t. The usual asymptotics simply fail.

2

The residual is the tell

A near-zero Durbin-Watson - a non-stationary residual - flags a spurious regression. Granger-Newbold: suspect it when R-squared exceeds DW.

3

Cointegration is the real thing

When a stationary combination exists, the levels genuinely belong together. Engle-Granger tests it by checking the residual.

Everything here is a calculator exercise: one spurious warning, one residual test.

How These Cases Work

Data, then arithmetic, then meaning

Each case has three steps

- Start from a regression of two $I(1)$ series.
- Read the residual - DW, then ADF.
- Then judge spurious or genuine.

What you should watch

- How R-squared exceeds DW in the trap.
- How the residual reveals the truth.
- How a stationary residual means cointegration.

The Spurious Trap

A great fit that means nothing

Regress one random walk on another, entirely unrelated one, and the output looks stellar: a high R-squared and a t-statistic that screams significance. But it's an illusion. The Durbin-Watson statistic, near zero, exposes it - the residuals are themselves a wandering random walk. We compute the warning and see the trap.

The regression output

- R-squared of 0.85.
- A t-statistic of 18.
- But a Durbin-Watson of just 0.15.

What we'll do by hand

- Apply the Granger-Newbold rule.
- Turn the DW into a residual autocorrelation.
- See why the fit is fake.

The Spurious Symptoms

Impressive numbers, no substance

two independent random walks $\rightarrow$ high R^2 , huge t – but the relationship is **spurious**

Why it happens

- Both series trend, so they seem to co-move.
- OLS reads shared drift as a relationship.
- The standard errors are badly wrong.

The giveaway

- The residuals don't settle - they wander.
- So the Durbin-Watson collapses toward zero.
- That's the signal to distrust the fit.

The Granger-Newbold Rule

When R-squared beats Durbin-Watson

Granger–Newbold: $R^2 = 0.85 > DW = 0.15 \Rightarrow$ suspect a spurious regression

The rule

- Compare R-squared to the Durbin-Watson.
- Here, 0.85 against 0.15.
- R-squared far exceeds DW.

The verdict

- That inequality is the classic red flag.
- Suspect a spurious regression.
- The high R-squared is not to be trusted.

What the DW Reveals

A residual near a unit root

$$DW \approx 2(1 - \hat{\rho}) \Rightarrow \hat{\rho} = 1 - \frac{0.15}{2} = \mathbf{0.925} \text{ (residuals } \approx \text{ a unit root)}$$

The conversion

- Durbin-Watson is about two times one minus rho.
- So rho is one minus DW over two.
- Here, 0.925.

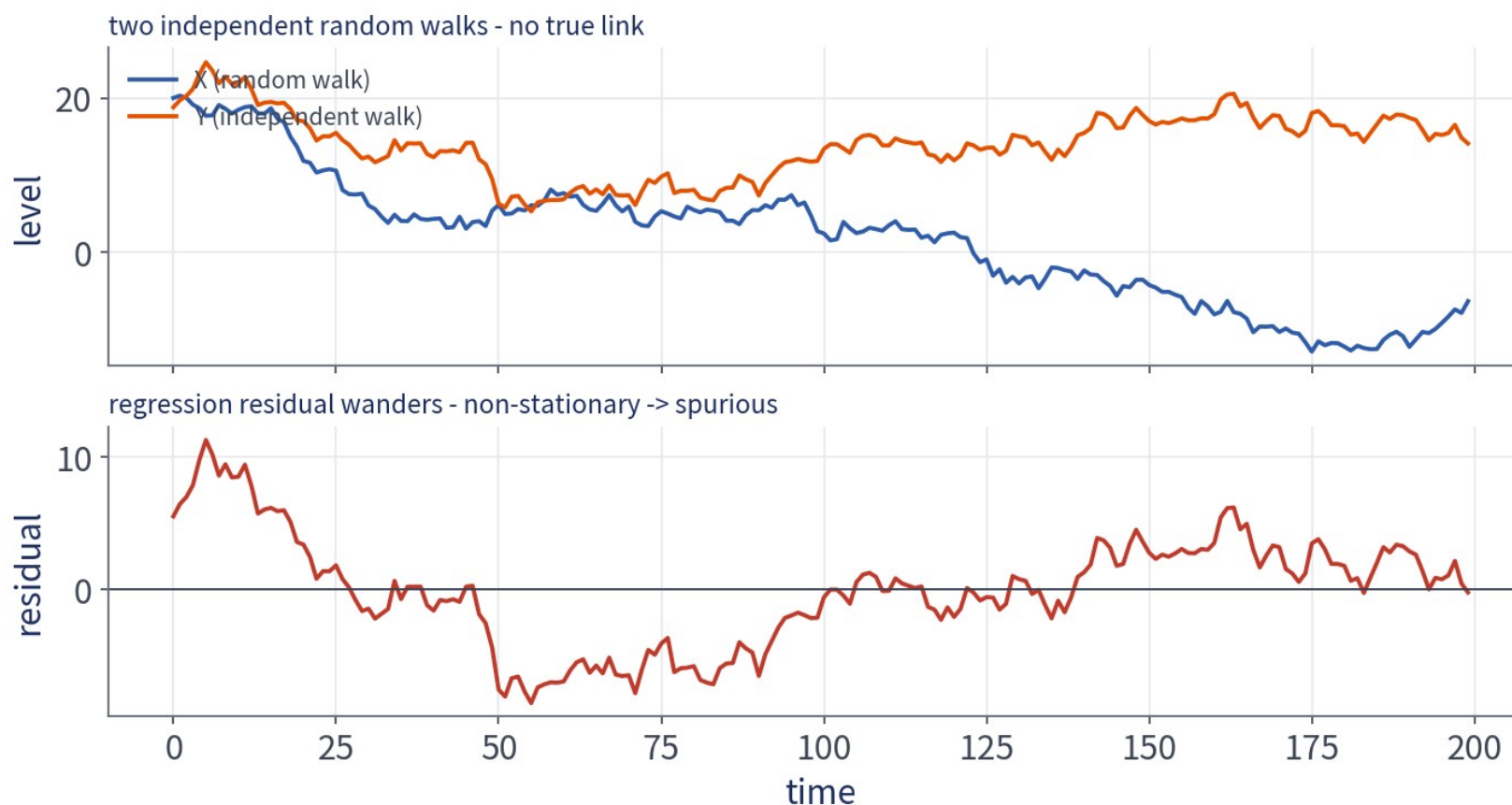
The meaning

- The residuals are almost perfectly persistent.
- That's essentially a unit root.
- Non-stationary residuals mean a fake link.

A Fit Built on Sand

The top panel shows two independent random walks that drift and appear to track each other - enough to give a high R-squared. The bottom panel shows the regression residual, which wanders without ever returning to zero. A non-stationary residual is the signature of a spurious regression: the relationship

Case A: high R-squared, but the residual never mean-reverts



Four Habits That Keep You Honest

Avoiding the trap

The discipline

- Check each series for a unit root first.
- Inspect the residual, not just the R-squared.
- Watch the Durbin-Watson.

The escape

- Difference to returns when there's no real link.
- But differencing throws away the long run.
- If the levels truly belong together, don't.

The Genuine Link

When the residual is stationary

Sometimes two non-stationary series really do move together - a stock and its hedge, two related futures. Then their combination, the spread, is stationary even though each series wanders. That's cointegration. The Engle-Granger test checks it by testing the residual for a unit root. We run it and find a real, tradeable

The genuine pair

- Two $I(1)$ series driven by a common trend.
- Regressing gives a slope of 1.05.
- The residual is the spread.

ib

What we'll do by hand

- Test the residual for stationarity.
- Compare to the Engle-Granger cutoff.
- Confirm cointegration.

Cointegration, Formally

Two wanderers, one stationary combination

$y_t, x_t \sim I(1)$ but $e_t = y_t - \beta x_t \sim I(0) \Rightarrow$ **cointegrated** (a stationary combination)

The definition

- Each series is integrated of order one.
- Yet some combination is stationary.
- That combination is the equilibrium.

Why it's special

- The levels share a long-run anchor.
- Differencing would discard exactly that.
- Cointegration keeps it.

The Engle-Granger Test

Test the residual for a unit root

$$\text{ADF on } \hat{e}_t : t = -4.20 < \underbrace{-3.34}_{\text{EG 5\%}} \Rightarrow \text{residual stationary} \Rightarrow \text{cointegrated}$$

The two steps

- First, regress one series on the other.
- Then run an ADF on the residual.
- A stationary residual means cointegration.

The special cutoff

- The residual is estimated, not observed.
- So the cutoff is -3.34, not -2.86.
- At -4.20, we clear it - cointegrated.

Spurious vs Cointegrated

The residual decides

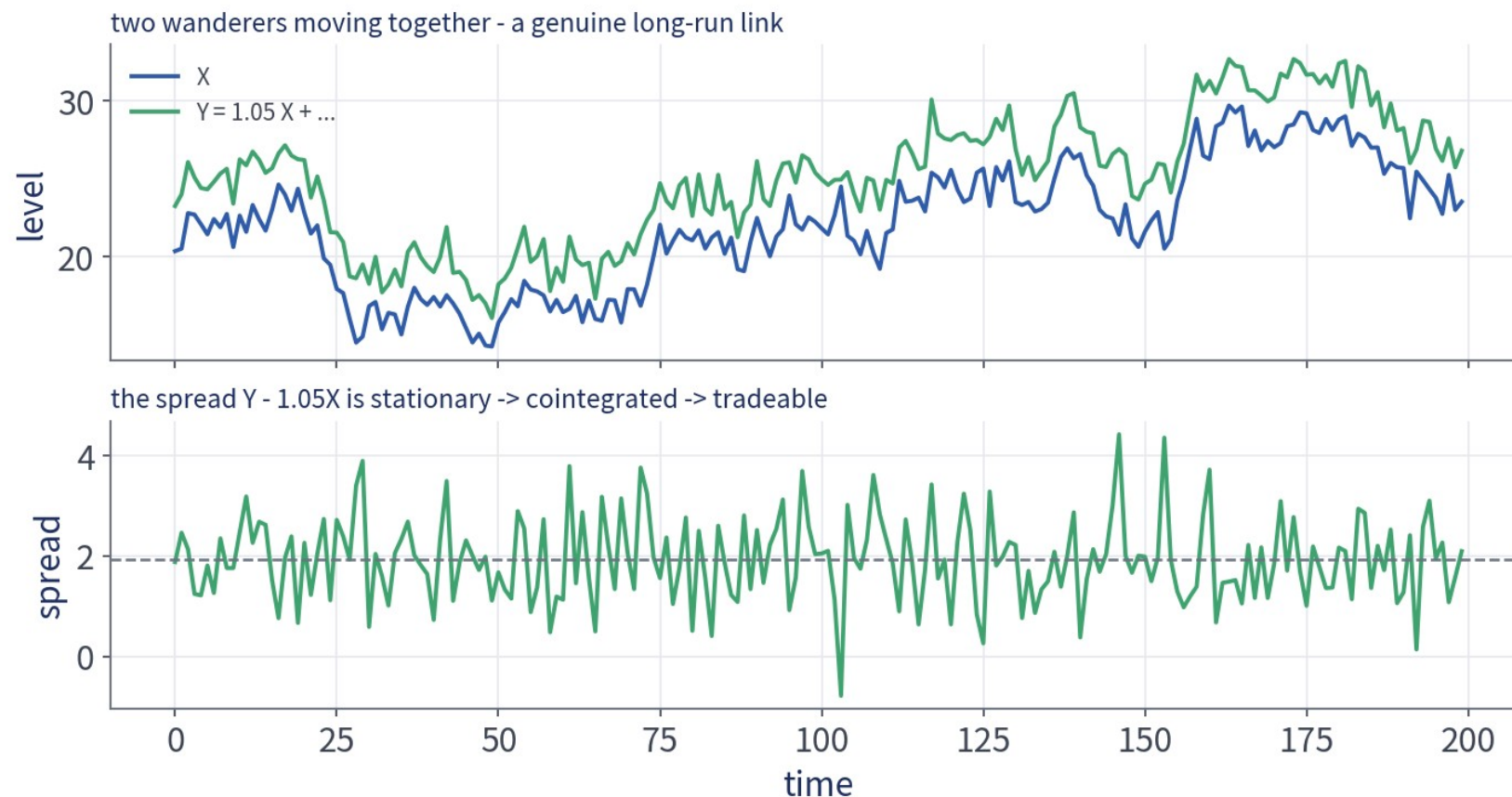
	spurious (Case A)	cointegrated (Case B)
Durbin-Watson	0.15 (near 0)	1.85 (healthy)
residual ADF	fails (~ -1.1)	rejects (-4.20)
residual is	non-stationary	stationary
the link is	fake	real, tradeable

Same-looking regressions, opposite truths - and the residual tells them apart. A wandering, non-stationary residual is spurious; a stationary, mean-reverting one is cointegration.

A Spread That Comes Home

The top panel shows two wandering series held together by a common trend - a genuine long-run link. The bottom panel shows their spread, which oscillates tightly around its mean instead of drifting. That stationary residual is cointegration, and its mean reversion is exactly what a pairs trade exploits.

Case B: the residual mean-reverts - a real equilibrium



From Equilibrium to a Trade

ECM and pairs trading

The error correction model

- When the spread strays, it gets pulled back.
- The ECM measures how fast.
- That pull is the tradeable signal.

Pairs trading

- Enter when the spread is far from its mean.
- Exit as it reverts.
- Cointegration is what makes it work.

The Cointegration Workflow

Suspect, test, then trade
The routine protects you and then pays you. When regressing non-stationary series, suspect a spurious result - check the Durbin-Watson and apply Granger-Newbold. Then test the residual properly: an Engle-Granger ADF that rejects means the residual is stationary and the series are cointegrated. A stationary spread is a real equilibrium - and mean reversion in that spread is what a pairs

The trap (Case A)

- R-squared 0.85 beats DW 0.15.
- Residual autocorrelation 0.925.
- Spurious - the link is fake.

Si The link (Case B)

- Residual ADF -4.20 clears -3.34.
- Stationary spread, DW 1.85.
- Cointegrated - and tradeable.

Reproduce the Trap and the Link

Direct Claude Code to check your arithmetic

GOAL Expose a spurious regression, then confirm a cointegrated pair.

[INPUT] Prompt · Claude Code

Write and run `part3_practice.py` that:

CASE A - spurious:

- * simulate two independent random walks, regress one on the other; print R-squared, t, and DW;
- * apply the R-squared > DW rule.

CASE B - cointegration:

- * simulate a cointegrated pair (common trend);
- * run the Engle-Granger test (regress, then ADF the residual) and compare to -3.34.

Then plot both pairs with their residual/spread.

Check against the slides

- spurious: $R^2 0.85 > DW 0.15$.
- $\rho \sim 0.925$ (near unit root).
- cointegrated: $ADF -4.20 < -3.34$.
- spread is stationary.

Deliverable: `part3_practice.py` whose printed numbers match the spurious warning and the EG test.

Three Questions on the Cases

If you can answer these, the practice has landed

Q1 What flags a spurious regression?

**A high R-squared
with a near-zero
Durbin-Watson**

Q2 What does cointegration require?

**A stationary
combination of non-
stationary series**

Q3 How does Engle-Granger confirm it?

**An ADF on the
residual that rejects
a unit root**

All three come from the Granger-Newbold rule, the definition of cointegration, and the residual test.

Three Pitfalls to Avoid

Where cointegration trips people

Common mistakes

- "A high R-squared means a real relationship."
- "Always difference to be safe."
- "Use standard ADF cutoffs on the residual."

The reality

- Check the residual - a low DW means spurious.
- Differencing discards a genuine long-run link.
- The residual is estimated - use -3.34, not -2.86.

Two Calculations to Remember

Part 3 applications in one screen

1

A high R-squared can be a lie

Two independent random walks gave an R-squared of 0.85 and a t of 18 - but a Durbin-Watson of 0.15. Since R-squared exceeds DW, Granger-Newbold flags it spurious; the implied residual autocorrelation of 0.925 confirms a near-unit-root residual.

2

A stationary residual is the real thing

For a genuine pair, the Engle-Granger ADF on the residual was -4.20, clearing the special -3.34 cutoff - so the residual is stationary and the series cointegrated. The Durbin-Watson of 1.85 agrees. The spread is a real equilibrium.

3

The residual is always the tell

Same-looking regressions, opposite truths: a wandering residual is spurious, a mean-reverting one is cointegration. That stationary spread is what the error correction model and pairs trading turn into a strategy.

Week 2 complete — from stationarity through unit roots to cointegration, the foundation every later model rests on.